

Advancing truss structure optimization—A multi-objective weighted average algorithm with enhanced convergence and diversity

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ABSTRACT

The challenge of achieving equilibrium between exploration and exploitation stands as a critical barrier in multi-objective metaheuristic optimization when applied to complex engineering problems such as truss structure design. The Multi-Objective Weighted Average Algorithm (MOWAA) presents a new methodology which employs adaptive weighted average position control to optimize population movement for enhanced solution quality. The performance evaluation of MOWAA relies on benchmarking it against five state-of-the-art multi-objective optimization algorithms NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME through eight truss structure optimization problems of increasing complexity. The evaluation of performance relies on three key metrics: Hypervolume (HV), Inverted Generational Distance (IGD) and Spacing (SP). MOWAA demonstrates superior performance compared to competing algorithms through its ability to generate Pareto fronts with higher HV values and lower IGD values and more uniform distribution. The enhanced performance of MOWAA demonstrates its superior capability to efficiently explore the objective space for finding optimal weight-minimization and compliance trade-offs. The robustness of MOWAA is proven through statistical validation with the Friedman rank test which establishes MOWAA as the leading approach with statistically significant advantages. MOWAA demonstrates runtime efficiency throughout truss optimization tasks of varying sizes which enables its practical application for real-world structural optimization problems. MOWAA emerges as a sophisticated and efficient optimization method which demonstrates strong capabilities for engineering applications and computational design.

1. Introduction

The increasing complexity of engineering problems raises the need for robust optimization algorithms that can efficiently solve difficult situations [1]. Traditional mathematical and numerical methods often deal with nonlinear constraints and search parameters, resulting in long calculation times and difficulty in obtaining accurate solutions. As a

result, metaheuristic optimization algorithms have become increasingly popular over the past two decades. Unlike traditional methods, these algorithms do not seek gradient information and can scan wider gaps, helping them avoid getting stuck in local values [2]. During the optimization process, metaheuristic algorithms iteratively adjust decision variables to achieve a goal, enabling effective navigation through complex environments—a very important capability in technology and

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other complex fields.

Metaheuristic algorithms are generally based on two main search strategies— Exploration and Exploitation [3]. Exploration directs algorithms to search the solution space globally, thus reducing the risk of falling into local optima. In contrast, exploitation focuses on improving nearby solutions to improve local solutions [4]. Balancing these ideas is crucial for algorithm optimization. Population-based metaheuristics like genetic algorithms (GAs) use specific operators and methods to integrate these ideas differently. Many metaheuristic algorithms have been developed and applied to many optimization problems in recent years. Research in this area is mainly focused on three methods—improvement of existing algorithms [5] creation of the most modern algorithms [6] and development of hybrid algorithms [7]. The efficiency together with adaptability of metaheuristic optimization algorithms has made them increasingly popular for solving complex engineering and real-world optimization problems. Research shows the Grey Wolf Optimizer (GWO) along with its variants demonstrate powerful capabilities in handling nonlinear and multimodal problems [8]. Engineering applications show enhanced performance with improved GWO variants including I-GWO because these variants improve both exploration and exploitation capabilities [8]. The Multi-Trial Vector-based Differential Evolution (MTDE) represents one of many DE-based approaches that aim to enhance optimization efficiency especially in constrained problems [9]. The Discrete Improved Grey Wolf Optimizer (DI-GWO) and the Effective Whale Optimization Algorithm (EWOA) demonstrate their adaptability to various problem domains through their hybrid and modified metaheuristic approaches for community detection and power system optimization respectively [10,11]. The Multi-Trial Vector-based Moth-Flame Optimization (MTV-MFO) and Multi-Trial Vector-based Sine Cosine Algorithm (MTV-SCA) represent advanced hybrid techniques that achieve better search efficiency and convergence stability [12] and [13]. This research develops an upgraded hybrid optimization framework that extends previous advancements to handle complex optimization challenges effectively.

The field of structural optimization represents an essential research domain because it helps designers create lightweight truss structures that maximize performance. The increasing use of multi-objective optimization frameworks helps solve these challenges by creating balanced solutions from competing design objectives. The Multi-Objective RIME Optimization Framework (MORIME) recently introduced shows promising results in improving truss structure design optimization efficiency and accuracy [14]. Space truss structure shape and sizing optimization benefits from powerful techniques based on cooperative coevolutionary-based algorithms which improve the exploration of complex design spaces [15]. The Leader Selection-based Multi-Objective Flow Direction Algorithm (MOFDA) demonstrates improved performance in solving multi-objective engineering design problems through enhanced convergence and solution diversity [16]. Recent advances in multi-objective optimization algorithms are development of cutting-edge algorithms like multi-objective heat transfer optimization (MOTEO) [17], dual-archive multi-objective multi-universe optimization (MOMVO2arc) [18], multi-objective hippo optimization (MOHO) [19]. These algorithms explain how to integrate advanced methods in engineering applications to achieve a balance between objectives. Hybrid algorithms use a weighted average to combine the advantages of multiple optimization methods. By calculating the weighted average of the solutions generated by different methods, these algorithms increase the ability to efficiently explore the search space, thereby reducing the probability of entering the optimal region. This approach increases the overall power and flexibility of the algorithm, resulting in optimal results for various problems.

The importance of hybrid optimization frameworks using machine learning techniques including deep learning and reinforcement learning and surrogate modelling has increased through recent studies to improve complex design optimization problems. Researchers apply hybrid optimization approaches by integrating evolutionary algorithms

with Kriging-based or Radial Basis Function (RBF) approximations to reduce compute time of high-dimensional multi-objective problem evaluations. These surrogate models operate as efficient prediction tools which reveal fitness landscapes to guide search processes by locating optimal areas within the solution space. Engineering optimization problems that require computational fluid dynamics (CFD) or finite element analysis (FEA) evaluations benefit significantly from surrogate model applications. Results from the combination of metaheuristic optimization techniques with machine learning algorithms including reinforcement learning-based exploration strategies have shown remarkable potential. The latest developments in Reinforcement Learning-Assisted Evolutionary Algorithms (RLA-EA) incorporate dynamic learning mechanisms that adjust exploration-exploitation balance through historical performance data analysis. The algorithm adaptive mechanism uses this ability to optimize search strategies which enhances both convergence speed and prevents getting stuck in local optimal solutions. Through deep learning-assisted metaheuristics neural networks extract optimization patterns from previous runs which enables accurate operator selection to guide searches toward potential Pareto-optimal solutions.

The "no free lunch" optimization approach [20] shows that a single optimization algorithm is not optimal for all problems. Metaheuristics are suitable for specific problem domains but not good at providing optimal solutions for other problem domains. This inconsistency often stems from a failure to strike the perfect balance between research and development. As a result, these can cause the algorithm to search well or fail at local optima. For this purpose, it is necessary to develop new algorithms to support different optimization problems.

Therefore, this paper uses the weighted average theory to develop a new metaheuristic optimization algorithm. In the proposed Multi-Objective Weighted Average Algorithm (MOWAA), the average score of all individuals at each time point is determined. Depending on the current situation, two different strategies are used to find the best solution. The first strategy consists of three distinct processes derived from the average score and the best solution obtained at each time point, allowing a complete analysis of the feasible search space. The second scheme builds on the first scheme and quickly converges to the best solution found each time to solve the search space. At the same time, it also performs the second function of reducing the probability of falling into the local optimum.

Multi-objective optimization techniques have achieved major progress which enhances computational performance when dealing with engineering problems especially structural design optimization. Current metaheuristic algorithms continue to encounter significant difficulties when trying to strike an equilibrium between exploration and exploitation. The commonly employed multi-objective optimization algorithms NSGA-II, MOEA/D and MORIME demonstrate poor performance on large-scale high-dimensional problems including truss optimization because they lack convergence speed and diversity maintenance abilities along with robustness. The optimization methods face difficulties in achieving both solution diversity preservation and quick Pareto-front convergence specifically for weight and compliance minimization objectives in structural engineering applications. Hybrid metaheuristic methods were developed to handle these issues through search strategy integration yet demand significant parameter adjustment alongside extensive computational power. Current hybrid optimization methods fail to maximize the potential benefits of weighted average-based search mechanisms because these mechanisms demonstrate unique capabilities in balancing exploration and exploitation during optimization processes. Current state-of-the-art multi-objective optimizers lack an organized weighted averaging framework which restricts their capacity to modify search approaches according to problem complexity and optimization advancement. These optimization methods achieve suboptimal results when applied to truss structures with many design variables and constraints thus demonstrating a requirement for better optimization methods. The proposed MOWAA addresses search challenges through a

weighted averaging method that directs search agents toward superior solution space areas systematically. The weighted average positioning technique in MOWAA enables candidate solutions to utilize global and local search algorithms which enables efficient early-stage exploration and strong convergence throughout subsequent iterations. The method successfully prevents premature convergence while sustaining multiple non-dominated solution options.

The research methodology uses a structured method to develop the algorithm and evaluate its performance along with statistical testing. The initial development phase of the MOWAA algorithm incorporates weighted average positioning as its fundamental search strategy. An adaptive function within the algorithm controls the balance between exploration and exploitation through its ability to adjust personal best and global best and weighted average position influences. The mathematical model presents comprehensive details about search process control through Levy flight exploration and adaptive weighting rules and feedback-driven information sharing methods. The second stage utilizes MOWAA to optimize multi-objective truss problems. Eight different truss structures, ranging from small-scale 10-bar trusses to complex 942-bar trusses, are optimized using MOWAA, with results compared against five well-established multi-objective optimization algorithms: NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME. Standard multi-objective performance indicators HV, IGD and SP evaluate the quality of Pareto-optimal solutions by assessing convergence along with distribution diversity and uniformity. The study third phase relies on statistical methods to confirm MOWAA operational effectiveness. The Friedman rank test serves as a statistical framework to determine algorithmic performance across truss design problems and confirms the observed improvements. The analysis of computational efficiency involves measuring runtime across different problem complexities to verify MOWAA ability to deliver performance at a competitive cost level.

The main purpose of this research is to establish MOWAA as a new multi-objective optimization algorithm which employs weighted averaging to achieve optimal exploration-exploitation balance. The proposed approach boosts solution convergence together with diversity and robustness specifically for large-scale structural optimization problems. MOWAA has an adaptive search mechanism that researchers believe will surpass current state-of-the-art metaheuristic algorithms for solving complex truss optimization problems. The research addresses multiple essential elements to reach its objective. The research develops a weighted average-based multi-objective optimization method that adapts search strategies through solution distribution analysis and problem complexity monitoring. The adaptive nature of MOWAA enables optimal resource distribution which enables both efficient Pareto front distribution and avoidance of premature convergence. The research evaluates MOWAA effectiveness through testing it against multiple benchmark multi-objective truss optimization problems which demonstrate its practical value for real-world structural engineering applications. MOWAA demonstrates its effectiveness by performing benchmark tests against NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME using HV, IGD and SP as performance metrics. These performance metrics evaluate how effectively the algorithm creates evenly distributed high-quality Pareto fronts which provide complete optimization capability assessments. A comprehensive analysis of MOWAA computational efficiency when solving problems of different complexities forms a vital part of this research. The evaluation examines MOWAA optimization capabilities through different structural optimization tasks which range from basic truss designs to complex systems containing hundreds of design variables. The study demonstrates MOWAA potential to scale up for practical engineering applications through these experiments. A Friedman rank test together with p -value analysis serves as the method for comprehensive statistical validation in this study. Statistical tests validate MOWAA superior performance regarding convergence quality and solution diversity which provides compelling evidence of its effectiveness as a multi-objective optimization

algorithm. This research adds value to structural optimization by developing a weighted average-based multi-objective optimization algorithm which effectively manages exploration and exploitation challenges. MOWAA demonstrates potential as a powerful tool for multi-objective truss optimization in engineering applications through its enhanced computational performance alongside superior solution quality and flexible capabilities for complex optimization problems. The MOWAA brings an innovative solution to multi-objective metaheuristic optimization fundamental challenge of exploration and exploitation balance. With its dynamic weighted average position strategy MOWAA enables candidate solutions to adapt their exploration of the search space while sustaining diversity alongside convergence. MOWAA demonstrates adaptive capabilities that distinguish it from traditional methodologies because static or semi-adaptive approaches struggle to strike a balance between exploration and exploitation. The main contribution of this study introduces a multi-objective optimization framework which utilizes weighted average position theory as its foundation. The standard metaheuristic optimization methods use direct solution perturbations together with fixed probability-based updates and heuristic strategies derived from natural phenomena. The position update system of MOWAA directs solution movement through a weighted average calculation of the best-performing candidates. The method simultaneously enhances local optimization and global exploration capabilities which produces a properly distributed Pareto front. The method accelerates convergence and guarantees diverse trade-off solutions on the Pareto front while being a fundamental requirement of multi-objective optimization. The work introduces Information Feedback Mechanism (IFM) as an integrated component of MOWAA to dynamically improve solution update procedures. The IFM uses historical best solutions as input while modifying the search path through continuous population performance evaluation. The IFM introduces an adaptive refinement process that improves search efficiency because it differs from existing algorithms which depend on predefined evolutionary operators or randomization techniques. The algorithm benefits from this approach by preventing itself from getting stuck in local optima which represents a typical issue in traditional evolutionary algorithms. The addition of this adaptive mechanism enhances algorithm stability while making it resilient to various optimization problems. The research presents a thorough benchmark evaluation of eight truss optimization problems ranging from simple 10-bar trusses to complex 942-bar structures. MOWAA undergoes rigorous benchmarking against five state-of-the-art multi-objective optimization algorithms including NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME through performance metrics HV, IGD and SP. The experimental data shows MOWAA delivers better convergence and diversity results than other algorithms especially when solving complex optimization problems with multiple constraints. The extensive benchmarking process has confirmed the algorithm effectiveness while proving its position as a leading solution method for structural optimization problems. This research introduces an optimized method to manage computational speed alongside solution quality. Multi-objective optimization faces an enduring challenge to find the right balance between solution quality and computational time requirements. High-performing algorithms demonstrate robust convergence capabilities through excessive computational demands which results in impractical usage for real-world applications operating under limited computational capacity. MOWAA demonstrates improved computational efficiency by reducing runtime while delivering high-quality Pareto fronts. The extensive runtime performance evaluation demonstrates MOWAA ability to effectively solve both smaller and larger scale truss optimization tasks. The study extends engineering optimization research by showing MOWAA effectively solves real-world structural design challenges. The development of multi-objective truss optimization problems with stress and weight and compliance limits presents an essential engineering design requirement. The application of MOWAA to these problems creates a practical optimization tool which engineers in aerospace, civil and mechanical fields can easily imple-

ment. The research methodology presents the basis for future developments in metaheuristic optimization to address large-scale high-dimensional and constrained multi-objective problems. This research established its fundamental importance through statistical testing which validated the algorithm performance using the Friedman rank test and statistical significance evaluation. The field of multi-objective optimization lacks numerous studies which provide statistical verification for their findings. Statistical analysis in this work confirms that the observed performance improvements are not random occurrences because they demonstrate statistical significance. The comparative analysis results yield low p -values which demonstrate MOWAA superior performance compared to other algorithms across solution quality metrics and distribution capabilities and computational efficiency metrics.

2. Methodology

2.1. Inspiration for the weighted average position

The main idea of the proposed weighted average algorithm (WAA) is to use weighted average position to solve the optimization problem. A weighted average is a statistic that measures the importance (or weight) of each item in a data set in calculating the overall average. Each element contribution to the average weight affects the same weight, meaning elements with a higher ratio significantly impact the final rating. This method is beneficial when different items in the data set have different values.

By applying weights in the calculation process, this method highlights the impact of specific elements, thus offering a more accurate representation of the overall average, thereby measuring the specific characteristics of the data set rather than the general average. The search field in each iteration controls the updates of individual locations in the next iteration. This new search field is created based on the current, personal and global best locations. Personal best (referred to as $X_{\text{Personal Best}}$) represents the best fitness value achieved by an individual in its history, global best (referred to as $X_{\text{Global Best}}$) represents the best fitness value achieved by the entire population. In the WAA algorithm, the weighted average represents the distribution of the current population. It combines the best personal and global positions to update the next position.

2.2. Mathematical model and algorithm

The Weighted Average Algorithm (WAA) is an innovative metaheuristic optimization technique that leverages the weighted average position of candidate solutions to navigate complex solution spaces effectively. Unlike traditional algorithms, WAA dynamically balances exploration and exploitation by calculating a weighted average that guides candidate movements toward optimal regions in each iteration. The algorithm employs exploitation strategies, which refine solutions around the global and personal best positions and exploration strategies, including Levy flights, to avoid premature convergence. This method is particularly suitable for multi-objective optimization, as it adapts its search path to optimize multiple conflicting objectives simultaneously, thereby ensuring robust performance across a wide range of optimization scenarios. The following mathematical model details each component of WAA, illustrating its ability to solve complex problems efficiently.

2.2.1. Initialization phase

The initialization phase in WAA defines the candidate solutions, where a population matrix X is generated with random values within specified boundaries. The position x_{ij} of each candidate in the population matrix X represents decision variables in the j -th dimension for the i -th candidate.

$$X = \{x_{ij} | i = 1, 2, \dots, N; j = 1, 2, \dots, n\} \quad (1)$$

where N is the Total number of candidate solutions (population size) and n is the dimensionality of the problem (number of variables).

Each candidate position is initialized using a uniform random distribution: $x_{ij} = \text{rand} \cdot (UB_j - LB_j) + LB_j$ where: rand is a random number between 0 and 1. LB_j and UB_j represent the lower and upper bounds for the j -th dimension.

2.2.2. Weighted average position calculation

The core feature of WAA is to calculate a weighted average position to guide the optimization process. The weighted average is computed based on the top-performing candidates (defined by their fitness values).

A subset of the best candidates, $N_{\text{Candidate}}$, is chosen based on their fitness values as per Eq. (2).

$$N_{\text{Candidate}} = \frac{(nP - 4) \cdot (it - 1)}{1 - \text{MaxIt}} + nP \quad (2)$$

where, nP : Initial population size; it : Current iteration number and MaxIt : Maximum number of iterations.

Compute the fitness values for each candidate and calculate the sum of fitnesses for the selected candidates as per Eq. (3).

$$\text{SumFitness} = \sum_{i=1}^{N_{\text{Candidate}}} \text{Fitness}(X_i) \quad (3)$$

The weighted average position, X_μ , is calculated as Eqs. (4) and (5).

For a "smaller-the-better" (STB) objective,

$$X_\mu = \frac{\sum_{i=1}^{N_{\text{Candidate}}} X_i \cdot (\text{SumFitness} - \text{Fitness}(X_i))}{\text{SumFitness} \cdot (N_{\text{Candidate}} - 1)} \quad (4)$$

For a "larger-the-better" (LTB) objective,

$$X_\mu = \frac{\sum_{i=1}^{N_{\text{Candidate}}} X_i \cdot \text{Fitness}(X_i)}{\text{SumFitness}} \quad (5)$$

Here, X_i is the position vector of the i -th candidate and $\text{Fitness}(X_i)$ is the fitness value of the i -th candidate.

2.2.3. Exploration and exploitation phases

The WAA balances **exploration** and **exploitation** through parameter-based decision-making in each iteration. The function controls this balance $f(it)$, which determines whether to explore or exploit based on the current iteration it and random factors as per Eq. (6).

$$f(it) = (\alpha \cdot \text{rand} - 1) \sin \left(\frac{\pi \cdot it}{\text{MaxIt}} \right) \quad (6)$$

If $f(it) \geq 0.5$, the algorithm enters **exploitation** mode and if $f(it) < 0.5$, the algorithm enters **exploration** mode.

Here, α is a control parameter for the exploration-exploitation balance and rand is a random value between 0 and 1.

■ Exploitation Strategies

WAA refines solutions in exploitation mode by focusing on areas near the personal best and global best positions, using three strategies.

First Strategy (Balanced Approach): Incorporates the weighted average position, personal best $X_{\text{PersonalBest}}$ and global best $X_{\text{GlobalBest}}$ as per Eq. (7).

$$X_i(it+1) = w_{11}(X_\mu(it) - X_{\text{GlobalBest}}(it)) + w_{12}(X_\mu(it) - X_{\text{PersonalBest}}(it)) + w_{13}X_\mu(it) \quad (7)$$

where w_{11}, w_{12}, w_{13} are random scalars between 0 and 1.

Second Strategy (Focus on Personal Best): This strategy emphasizes movement toward the personal best position, as per Eq. (8).

$$X_i(it+1) = w_{21}(X_\mu(it) - X_{\text{PersonalBest}}(it)) + w_{22}X_{\text{PersonalBest}}(it) \quad (8)$$

where w_{21}, w_{22} are random scalars between 0 and 1.

Third Strategy (Focus on Global Best): This strategy directs movement toward the global best position, as per Eq. (9).

$$X_i(it+1) = w_{31}(X_{\mu}(it) - X_{\text{GlobalBest}}(it)) + w_{32}X_{\text{GlobalBest}}(it) \quad (9)$$

where w_{31}, w_{32} are random scalars between 0 and 1.

■ Exploration Strategies

WAA leverages strategies to search the broader solution space in exploration mode to avoid local optima.

Levy Flight Strategy: This random walk approach helps candidates explore the search space dynamically as per Eq. (10).

$$X_{ij}(it+1) = X_{\text{GlobalBest},j}(it) + S \quad (10)$$

where S is the step length calculated from a Levy distribution as per Eq. (11).

$$S = \frac{U}{|V|^{1/\beta}} \quad (11)$$

Here, U and V are random values drawn from a normal distribution and β is a parameter controlling step length.

Boundary Exploration Strategy: For cases where Levy flight might converge prematurely, a boundary-driven exploration step adjusts the search space: $X_i(it+1) = \text{rand} \cdot (UB_{\min} - LB_{\min}) + LB_{\min}$ where $UB_{\min}$ and $LB_{\min}$ are the minimum bounds across all dimensions.

This WAA framework integrates weighted averaging, adaptive exploration-exploitation switching and structured movement strategies to efficiently navigate complex solution spaces. The balance between convergence speed and global optimality makes WAA robust to various optimization challenges.

2.3. Multi-objective weighted average optimization algorithm

MOWAA functions as a multi-objective extension of Weighted Average Algorithm (WAA) which maintains weighted average-based position updates yet implements specialized mechanisms for multi-objective optimization. MOWAA develops from WAA fundamentals to provide essential updates which solve multi-objective optimization problems by enabling trade-off exploration and solution diversity and efficient convergence. The essential difference between WAA and MOWAA emerges from their separate methods for problem resolution. The optimization framework of WAA addresses single-objective problems because it finds a specific optimal outcome through weighted position averaging tactics. MOWAA builds upon weighted average positioning by implementing diverse mechanisms that utilize Pareto-based ranking and non-dominated sorting and diversity preservation strategies across multi-objective domains. MOWAA achieves improved solution distribution through its added features which prevent it from focusing on a single optimal outcome. MOWAA implements a weighted average position update mechanism that addresses multiple conflicting objectives as its primary modification. MOWAA deploys an adaptive solution update method which evaluates solution dominance relationships based on Pareto principles instead of using single-objective best performer criteria. Through trade-off guidance the search process maintains direction which enables the algorithm to explore the objective space effectively without getting stuck on a single solution. The Information Feedback Mechanism (IFM) of MOWAA dynamically modifies candidate solution weight coefficients using their multi-objective fitness rankings. MOWAA uses this mechanism to maintain adaptive balance between solution convergence and diversity thus keeping high-quality non-dominated solutions while preventing solution clustering. The IFM strategy makes MOWAA more robust because it enables dynamic adjustments of search behavior which results in better performance

across challenging optimization landscapes. MOWAA incorporates crowding distance-based selection as an extension to maintain uniform solution distribution across the Pareto front. The crowding distance mechanism in MOWAA actively preserves solution diversity because WAA lacks this capability. The adaptation plays a vital role in obtaining properly distributed Pareto fronts in multi-objective optimization because it stops solutions from clustering in particular search space areas. MOWAA implements diverse exploration strategies together with specialized exploitation approaches designed for multi-objective problem spaces. The implementation of Lévy flight-based exploration enables broad global search while avoiding premature convergence through distant solution jumps. Solution diversity improves through boundary exploration strategies which discover extreme trade-off solutions during the search process. The mechanisms prove essential for engineering applications because they enable discovery of vital diverse Pareto-optimal solutions needed for decision-making. MOWAA represents WAA principles for multi-objective optimization through the utilization of Pareto-based selection and adaptive weighting strategies and crowding distance mechanisms and advanced exploration-exploitation strategies.

The Crowding Distance (CD) mechanism helps achieve a well-distributed Pareto front through its dual function of solution diversity maintenance and cluster prevention. The MOWAA combines CD with weighted average-based offspring generation and Information Feedback Mechanism (IFM) to achieve proper convergence-diversity equilibrium. The MOWAA framework performs a continuous cycle of population selection and refinement during its operational process. The weighted average strategy positions solutions at the search space center by combining both population distribution information with top-performing individual solutions. The search direction receives guidance through a weighted centroid calculation mechanism that evolves with each new solution produced. Any multi-objective optimization process faces a critical challenge because it tends to converge prematurely toward restricted objective space areas. The CD mechanism addresses this challenge by improving diversity without compromising solution quality. The CD mechanism operates during selection after the weighted average approach generates offspring populations. The combined population receives non-domination level-based sorting which places non-dominated solutions into the first level. The selection process determines the next generation members by computing CD values within individual non-dominated fronts. The CD value of each individual represents its solution density in relation to its surrounding area. The algorithm determines solution quality through objective distance calculations between neighboring solutions which are added to form a complete distance score across all objectives. The selection process favors solutions that have elevated CD values thus sustaining population diversity across the Pareto front. Information Feedback Mechanism (IFM) boosts the selection probability adjustment through fitness value calculations which improve overall balance in the population. The integration of IFM with CD enables solutions to be selected according to their dominance rank as well as their distribution. Through the IFM process promising solutions with high fitness values guide the next generation population while CD avoids population clustering at local optima. MOWAA achieves both solution distribution maintenance and Pareto front convergence acceleration through this dual mechanism. MOWAA obtains optimal exploration versus exploitation performance through the coordinated operation of its components. The weighted average approach delivers adaptive global search capabilities while intermediate fitness manipulation improves promising solution refinement power and diversity conservation attributes derive from competitive diversity. The combined approach creates an algorithm which demonstrates robust performance and adaptability while effectively tackling complex multi-objective optimization challenges.

The MOWAA introduces the Crowding Distance (CD) mechanism to weighted average-based offspring populations which establishes this method as a major advancement in multi-objective optimization over

conventional selection approaches. This approach stands out because it controls population diversity dynamically and directs searches toward distributed Pareto-optimal solutions. NSGA-II and MOEA/D together with other traditional multi-objective optimization algorithms use dominance ranking and CD-based selection as their primary mechanisms. These solution diversity maintenance methods succeed at their task yet they encounter premature convergence difficulties and lack control over population movement. The MOWAA framework presents a core advancement through its integration of CD with weighted average-based approaches which calculate offspring positions through adaptive centroid computations. The computation uses best solutions together with population distribution data to guide MOWAA through complex search areas. The implementation of CD with weighted average-based offspring populations produces two major benefits. Weighted averaging acts as an automatic stabilizer for the search procedure because it reduces outlier solution impacts which enables population characteristics to guide offspring positioning instead of individual biases. The optimization process becomes more resilient through this feature because it stops erratic search behavior which traditional evolutionary algorithms frequently display. The CD mechanism protects the selection process from losing diverse solutions by preventing weighted averaging from grouping solutions near the centroid. MOWAA achieves optimal distribution of solutions across the Pareto front through CD integration which maintains diversity while maintaining convergence. This approach offers the advantage of automatic exploration and exploitation tendency adjustment throughout its execution. Standard selection methods need predefined parameters together with heuristic adjustments to balance diversity against convergence objectives. MOWAA weighted average approach dynamically responds to problem landscape changes and CD enhances selection through solution spacing maintenance. Because it adapts automatically MOWAA requires less human intervention to set parameters thus improving its performance for problems of different complexity levels. The CD mechanism within MOWAA provides exceptional benefits when dealing with high-dimensional multi-objective problems. Large-scale optimization situations with growing numbers of objectives make diversity maintenance significantly harder. The weighted average approach enables efficient population movement guidance while the Crowding Distance mechanism stops solutions from focusing in specific objective space areas. MOWAA dual-layer approach surpasses standard selection-based algorithms by producing both a diverse and high-quality Pareto front. The combination of CD with weighted average-based offspring population creation establishes a new approach for multi-objective optimization. MOWAA utilizes weighted averaging alongside CD to obtain superior diversity control and enhanced convergence stability and greater adaptability across different optimization scenarios. MOWAA demonstrates robust scalability as a solution for complex multi-objective engineering design problems and shows particular strength in structural optimization applications.

The MOWAA algorithm starts with a random population. current generation is t , a_i^t and a_i^{t+1} the i -th individual at t and $(t+1)$ generation. u_i^{t+1} the i -th individual at the $(t+1)$ generation generated through the WAA algorithm and parent population P_t . the fitness value of u_i^{t+1} is f_i^{t+1} and U^{t+1} is the set of u_i^{t+1} . Then, we can calculate x_i^{t+1} according to u_i^{t+1} generated through the WAA algorithm and Information Feedback Mechanism (IFM) Eq. (12)

$$x_i^{t+1} = \delta_1 u_i^{t+1} + \delta_2 a_i^t; \delta_1 = \frac{f_k^t}{f_i^{t+1} + f_k^t}, \delta_2 = \frac{f_i^{t+1}}{f_i^{t+1} + f_k^t}, \delta_1 + \delta_2 = 1 \quad (12)$$

where x_k^t is the k th individual we chose from the t th generation, the fitness value of x_k^t is f_k^t , δ_1 and δ_2 are weight coefficients. Generate offspring population Q_t . Q_t is the set of a_i^{t+1} . The combined population $R_t = P_t \cup Q_t$ is sorted into different w-non-dominant levels ($F_1, F_2, \dots, F_l, \dots, F_w$). Begin from F_1 , all individuals in level 1 to l are added to $S_t = \bigcup_{i=1}^l F_i$

and remaining members of R_t are rejected. If $|S_t| = N$ no other actions are required and the next generation is begun with $P_{t+1} = S_t$ directly. Otherwise, solutions in S_t are included in P_{t+1} and the remaining solutions $N - \sum_{i=0}^{l-1} |F_i|$ are selected from F_l according to the Crowding Distance (CD) mechanism, the way to select solutions is according to the CD of solutions in F_l . The larger the crowding distance, the higher the probability of selection and check termination condition is met. If the termination condition is not satisfied, $t = t + 1$ then repeat and if it is satisfied, P_{t+1} is generated, it is then applied to generate a new population Q_{t+1} by WAA algorithm shown in Fig. 1. Such a careful selection strategy is found to cause computational complexity of M -Objectives $O(N^2M)$.

Pseudo-code of MOWAA algorithm:

Generation t of MOWAA algorithm with IFM procedure

Input: Initial population P_t ($t = 0$), population size (N), WAA algorithm parameters

Output: P_{t+1}

- 1: $S_t = \emptyset, i = 1$
- 2: **Proposed Information Feedback Mechanism (IFM)**
WAA algorithm applies on the initial population P_t to generate u_i^{t+1} ,
Calculate $x_i^{t+1} = \delta_1 u_i^{t+1} + \delta_2 a_i^t$; $\delta_1 = \frac{f_k^t}{f_i^{t+1} + f_k^t}$; $\delta_2 = \frac{f_i^{t+1}}{f_i^{t+1} + f_k^t}$, $\delta_1 + \delta_2 = 1$
 $Q_t = \text{WAA algorithm } x_i^{t+1} \cup Q_t$; (Q_t is the set of x_i^{t+1})
- 3: $R_t = P_t \cup Q_t$
- 4: Different non-dominated levels ($F_1, F_2, \dots, F_l$) = non-dominated sort (R_t)
- 5: **Repeat**
- 6: $S_t = S_t \cup F_i$ and $i = i + 1$
- 7: **Until** $|S_t| \geq N$
- 8: Last front to be included $f^l = U_{i=1}^l F_i$
- 9: **if** $|S_t| \geq N$ **then**
- 10: $P_{t+1} = S_t$ **Elseif** $P_{t+1} = \frac{S_t}{F_l}$
- 11: **Else**
- 12: $P_{t+1} = U_{i=1}^{l-1} F_i$
- 13: Chosen $N - |U_{i=1}^{l-1} F_i|$ individuals are selected from f_l according to the large value of
 $\text{Crowding Distance } (CD^i) = \frac{f^{i+1} - f^{i-1}}{f^{\max} - f^{\min}}$
- 14: **end if**

MOWAA emerged from a complete examination of weighted-based optimization methods to overcome fundamental multi-objective optimization challenges in engineering applications including truss structure design. The innovation of MOWAA faces questions because weighted particle swarm optimization (PSO) variants alongside other weight-based techniques already dominate metaheuristic algorithms. MOWAA presents multiple essential breakthroughs which set it apart from traditional weighted-based techniques thus making it important for the field. The main distinguishing element of MOWAA involves its improved weighted average calculation system. MOWAA outperforms traditional weighted PSO variants through its dynamic selection of promising candidates which receive adaptive weight assignments based on fitness rather than static weight distributions. The optimization method uses local and global data together with diversity-preserving techniques to achieve balanced exploration and prevent premature convergence. The adaptive switching system of MOWAA controls the transitions between exploration and exploitation phases. The exploration strategies of numerous conventional weighted optimization techniques including PSO and its variants lead to premature convergence issues. MOWAA resolves this drawback through a structured parameterized function that automatically chooses between solution space expansion and high-quality solution refinement. MOWAA demonstrates better search performance than weighted approaches because its adaptive design automatically controls exploration and exploitation phases instead of requiring manual adjustment. Information Feedback Mechanism (IFM) represents a key MOWAA innovation because it uses historical population data to improve candidate solutions. MOWAA Information Feedback Mechanism systematically incorporates past

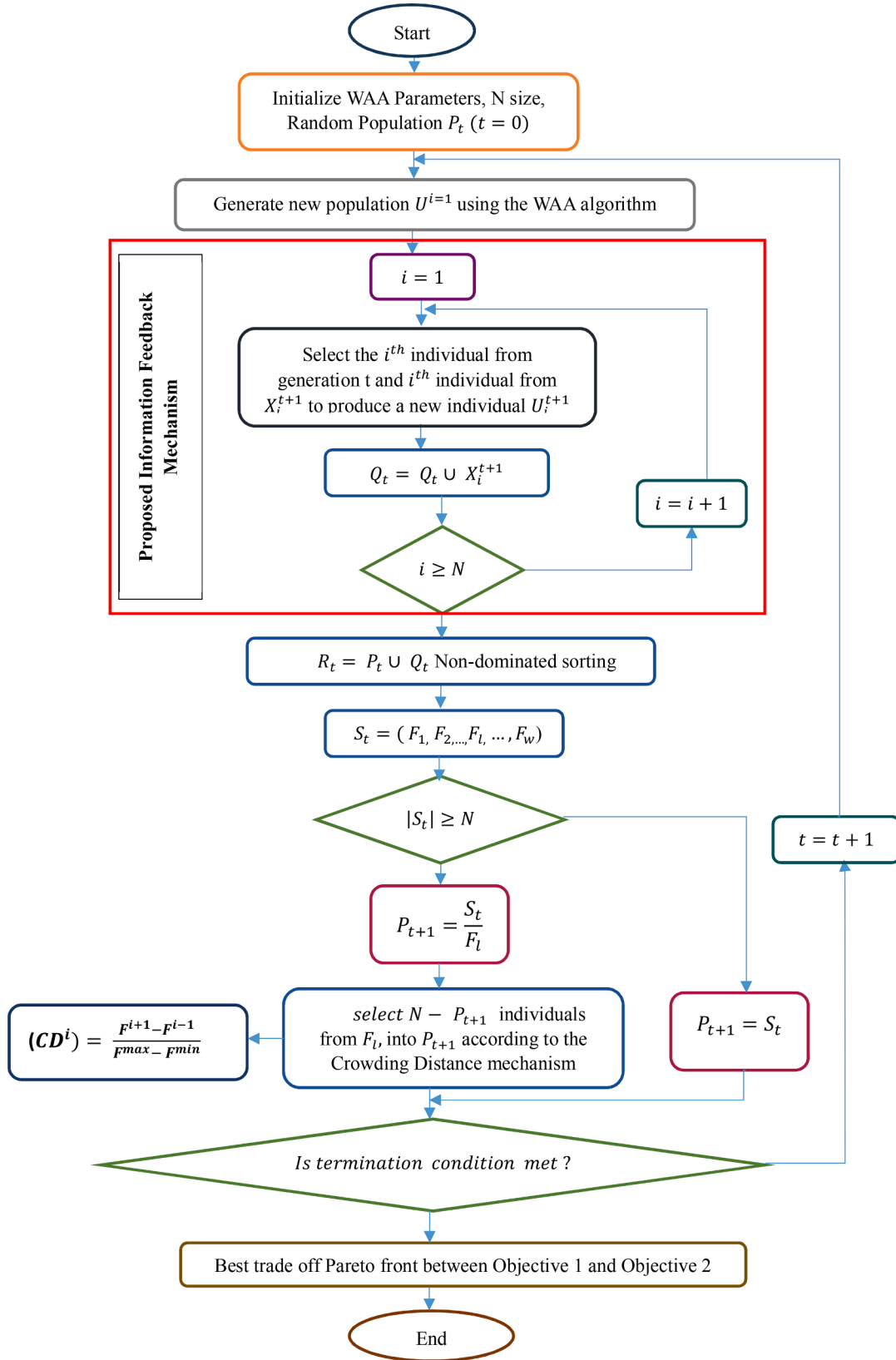


Fig. 1. Flow chart of MOWAA algorithm.

population performance to refine search directions through continuous updates of previously successful solutions while weighted PSO variants depend on direct velocity-position updates without this systematic integration. The mechanism improves both optimization convergence

and population diversity at the same time to create a stronger optimization process. MOWAA serves specifically as a multi-objective optimization tool whereas traditional weighted PSO methods were originally built to address single-objective problems. Thru the inclusion of Pareto

dominance principles MOWAA efficiently controls contradictory objectives by making dynamic trade-offs between multiple priorities without strict factor weighting rules. The non-dominated sorting approach in MOWAA generates diverse and well-distributed Pareto fronts while solving the weighting constraint requirements that limit weighted PSO approaches. MOWAA implements hypervolume-guided selection which standard weighted optimization algorithms typically lack. MOWAA achieves optimal Pareto front distribution through its integration of weighted average positioning with hypervolume maximization techniques to expand objective space coverage. The algorithm ability to detect optimal trade-offs between multiple conflicting objectives becomes more powerful through this enhancement leading to superior performance in multi-objective optimization. MOWAA distinguishes itself from classical weighted-based methods through its mathematical modeling which produces a strong weighted average position updating equation. MOWAA enables candidate solutions to explore different trajectories through its combination of balanced updating with personal best-focused and global best-focused strategies. The mathematical framework behind this algorithm enables its effective application to large-scale truss structure optimization problems that conventional weighted PSO variants cannot efficiently solve. The diversity-preserving strategy used in MOWAA prevents solutions from getting trapped in local optimal regions while simultaneously improving both solution diversity and computational efficiency. Through weight-based averaging the algorithm guides each update cycle toward wide-ranging candidate solutions while reducing the chance of becoming trapped in local optima. The numerical efficiency of MOWAA results from its structured selection methods which minimize redundant function evaluations.

3. Formulating the truss design problem

3.1. Multi-objective truss design problem

The multi-objective truss optimization problem is complex because of the conflicting goals, intricate constraints and discrete variables related to the cross-sectional areas as per Eq. (12). This paper aims to address this problem by minimizing the total weight and compliance while adhering to the specified stress constraints. [21]. The conflicting objectives cannot be simultaneously achieved to maximize overall value as per Eq. (13). Furthermore, the nonlinear stress constraints also restrict the range of design variables, resulting in more complex optimal design problems. Unlike the single-objective problem, the multi-objective truss optimization problem seeks to find the Pareto solution set, which consists of all feasible solutions that are not dominated by any other solution rather than a single optimal solution. The structure of a multi-objective truss optimization can be expressed as:

$$\text{Find } X = \{x_1, x_2, x_3, \dots, x_m\} \quad (12a)$$

$$\text{Min } f_1(x) = \sum_{i=1}^m A_i \rho_i L_i \text{ and Min } f_2(x) = \text{compliance} = u^T F \quad (13)$$

$$\text{Subject to constraints : } |\sigma_i| - \sigma_{\max} \leq 0 \text{ and } A_i^{\min} \leq A_i \leq A_i^{\max} \quad (14)$$

where A_i = design variable of the cross-sectional area for i^{th} element, m = number of design variables, f_1 and f_2 = structural mass and compliance, respectively, ρ_i and L_i = mass density and length of the i^{th} element, respectively, σ_i and $\sigma_{\max}$ = stress and the allowable value of the i^{th} element and $A_i^{\min}$ and $A_i^{\max}$ = lower and upper bounds of cross-sectional areas of design variables as per Eq. (14).

3.2. Formulation and properties of different truss structures

This research explores eight multi-objective truss sizing optimization

cases— 10-bar, 25-bar, 37-bar, 60-bar, 72-bar, 120-bar, 200-bar and 942-bar trusses. The main objectives are to reduce the weight of the structure and its compliance while meeting stress constraints. The optimization utilizes weight and compliance equations, with displacement and loading vectors derived from finite element analysis. Figs. 2-19 illustrates the truss ground structures for each case.

3.2.1. 10-Bar planer truss

The 10-bar truss problem (Fig.2) uses cross-sectional areas as design variables (A_i for $i = 1, 2, \dots, 10$). The stress constraint is set at $\sigma_{\max} = 400$ MPa, with material density $\rho = 7850$ kg/m³ and Young's modulus $E = 200$ GPa. The size variables range between $[1, 1.5, 2, \dots, 21] \times 10^{-3}$ m². The loading condition involves downward forces of 1000 KN applied at nodes 2 and 4.

3.2.2. 25 bar 3D truss

This truss, demonstrated in Fig.3, also adheres to a stress constraint of $\sigma_{\max} = 400$ MPa and uses the same density and Young's modulus as the 10-bar truss. The design variables are similarly defined, but the loading condition is more complex, involving multiple forces at various nodes. This structure requires optimization to handle the increased loading points while minimizing weight and compliance.

3.2.3. 37-Bar planer truss

The 37-bar truss (Fig. 4) shares the same material properties and stress limits as the 10-bar truss. However, it introduces multiple loading cases with 1000 KN forces. The design variables remain in the same range.

3.2.4. 60-Bar ring truss

The 60-bar ring truss (Fig.5) follows the same material and stress constraints, with design variables within $[1, 1.5, 2, \dots, 21] \times 10^{-3}$ m². Multiple load cases are applied across different nodes, making optimization for both weight and stress distribution essential.

3.2.5. 72-Bar tower 3-D truss

The 72-bar truss (Fig. 6) maintains the same material properties and stress constraints as previous cases. Its design variables follow the same range and the truss is subject to two loading conditions, requiring high optimization due to its complex force distribution.

3.2.6. 120 bar truss

This truss continues with the same constraints of stress, density and Young's modulus. However, the design variables S has a range of $[1, 1.5, 2, \dots, 21] \times 10^{-3}$ m². The structure faces various loading conditions involving different force magnitudes at nodes, which requires careful optimization to ensure structural performance under heavy loads. The 120-bar dome truss ground structures are indicated in Fig. 7.

3.2.7. 200-Bar truss

The 200-bar truss (Fig.8) adheres to the same stress, density and Young's modulus constraints as other trusses. This structure faces lateral and vertical loads at multiple nodes, increasing the complexity of the optimization problem.

3.2.8. 942-Bar tower truss

The 942-bar tower truss, depicted in Fig. 9, is a highly complex structure subjected to various vertical and lateral loading conditions. Like the other trusses, it follows the same material constraints. The cross-sectional areas, denoted as A_i for $i = 1, 2, \dots, 942$, are chosen from a set of discrete values. The significant size and complexity of this structure require a broader range of cross-sectional areas to accommodate the substantial forces. The load cases are divided into sections, including vertical loadings of $P_z = -100$ KN and lateral loadings of $P_x = \pm 3$ KN, which together ensure that the structure can withstand both horizontal and vertical forces.

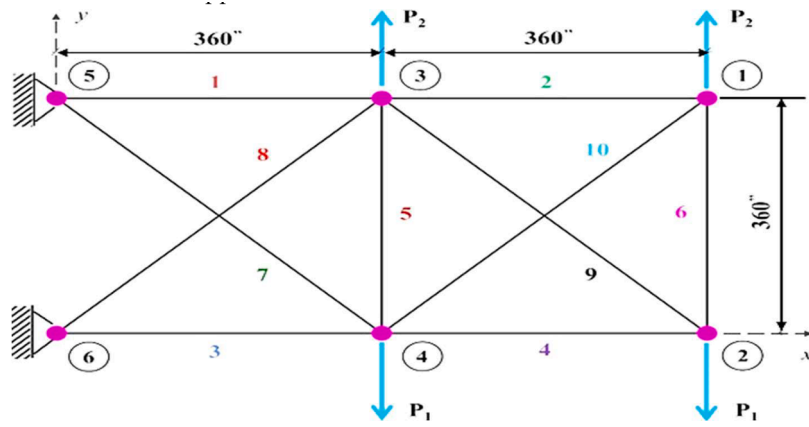


Fig. 2. 10 bar Truss.

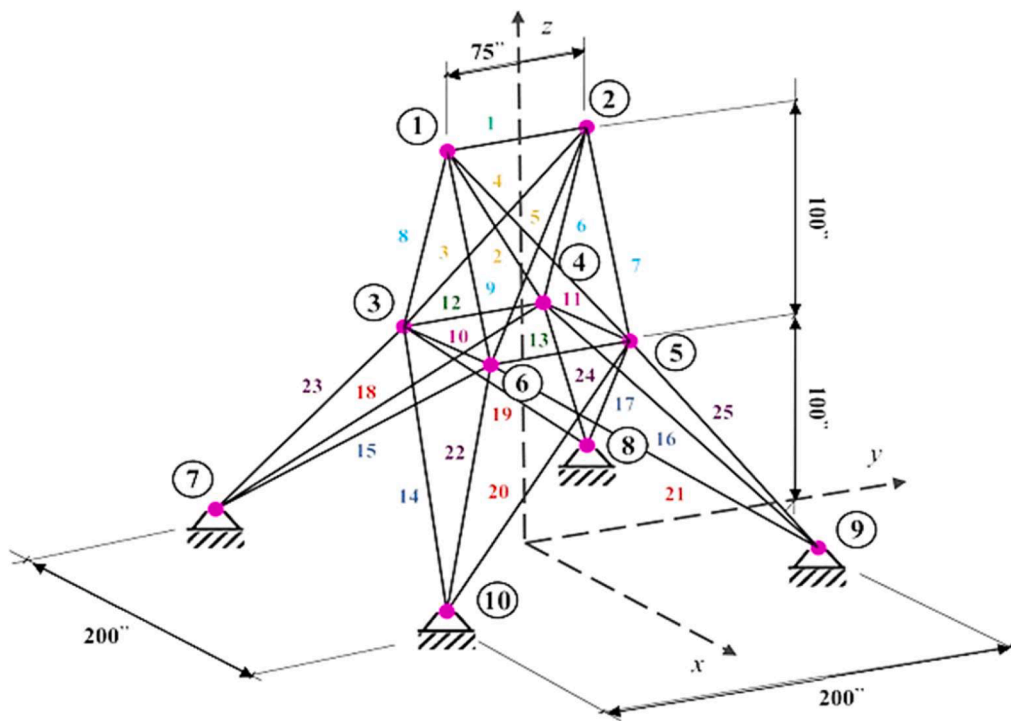


Fig. 3. 25 bar Truss.

4. Results and discussion

4.1. Convergence and diversity analysis

State-of-the-art multi-objective optimization algorithms face their main challenges when achieving effective convergence-diversity balance. Different existing multi-objective optimization algorithms including NSGA-II and MOEA/D and MOLCA and MOEDO and MORIME demonstrate unique features which determine their performance and stability when applied to truss optimization problems. A comprehensive evaluation of current multi-objective optimization algorithms reveals their shortcomings while demonstrating how the MOWAA addresses these weaknesses to enhance optimization results.

NSGA-II represents a popular multi-objective evolutionary algorithm yet it presents three fundamental drawbacks including early convergence behavior and limited diversity in high-dimensional spaces together with difficulties managing complex Pareto fronts. NSGA-II dominance-based selection algorithm creates strong concentration on

specific solution domains which reduces the algorithm capacity to properly explore various approaches. The crowding distance calculations in NSGA-II sometimes produce uneven solution distribution along the Pareto front which causes diversity issues in highly constrained problems like truss optimization. MOWAA solves these issues through its weighted average position strategy which automatically controls exploration against exploitation. MOWAA implements a fitness-based information feedback system to achieve effective solution convergence and diversity control beyond the dominance ranking and crowding distance approach used in NSGA-II. The method prevents solutions from clustering in specific areas while producing a balanced Pareto front across multiple truss optimization problems.

The MOEA/D framework divides multi-objective optimization problems into separate scalar optimization subproblems which it solves simultaneously. The method delivers enhanced computational performance yet encounters difficulties sustaining diverse solutions especially when working with complex front distributions that are unevenly distributed. Weight vector selection in MOEA/D remains its primary

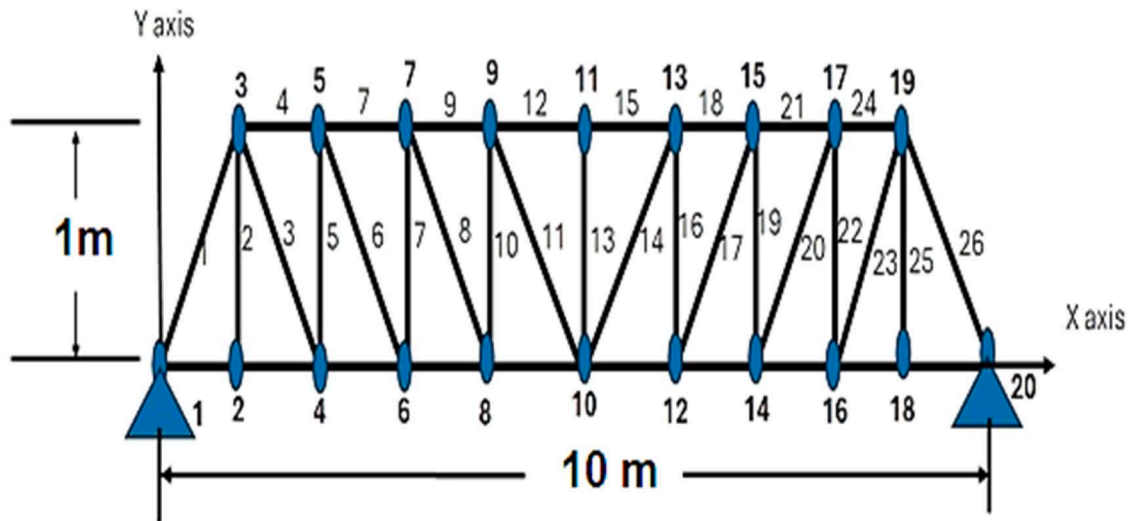


Fig. 4. 37 bar Truss.

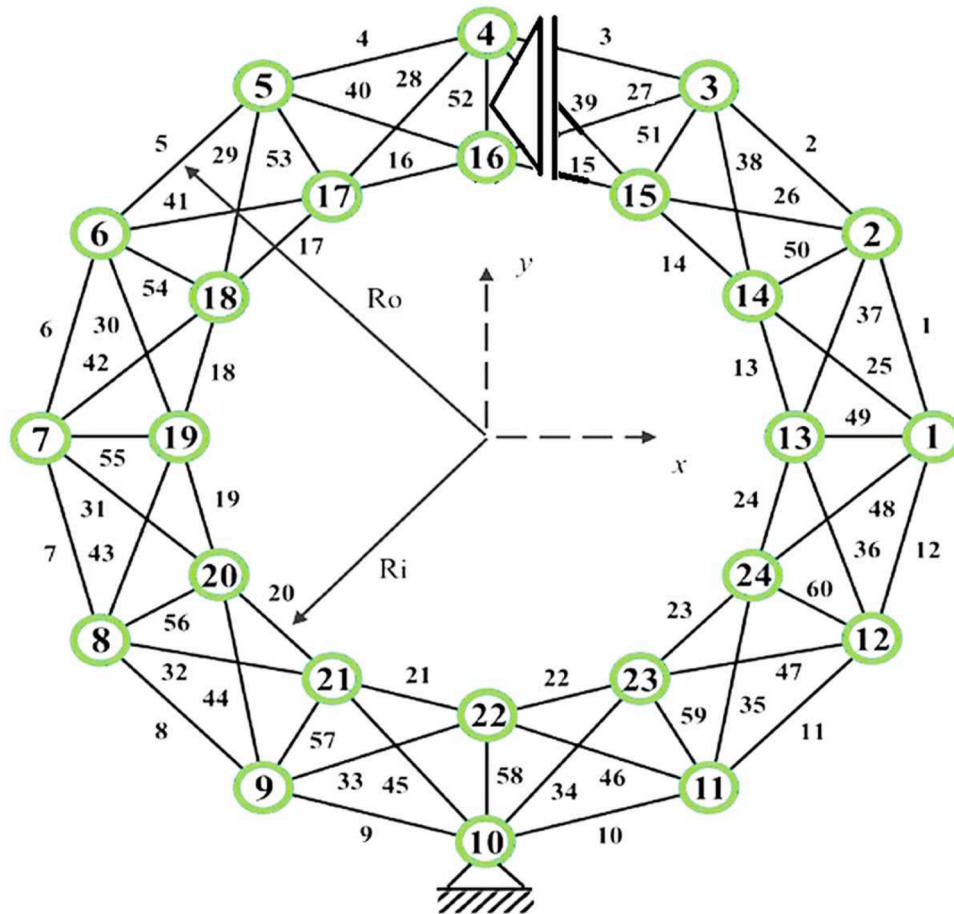


Fig. 5. 60 bar Truss.

weakness because it produces suboptimal convergence results for irregular Pareto fronts. The fixed neighborhood structure of MOEA/D restricts its capability to escape local optima and reduces its effectiveness when solving high-dimensional truss optimization problems. MOWAA solves these problems through its adaptive weighted average method which changes search strategies according to population distribution patterns. The adaptive search strategy mechanism in MOWAA

helps solutions avoid local optima while improving their distribution throughout the objective space. A dynamic exploration-exploitation parameter maintains both convergence and diversity across the Pareto front during optimization.

MOLCA fixed learning framework reduces its ability to adjust when optimization landscapes change. The method produces gradual convergence speed when tackling problems with complex curved or

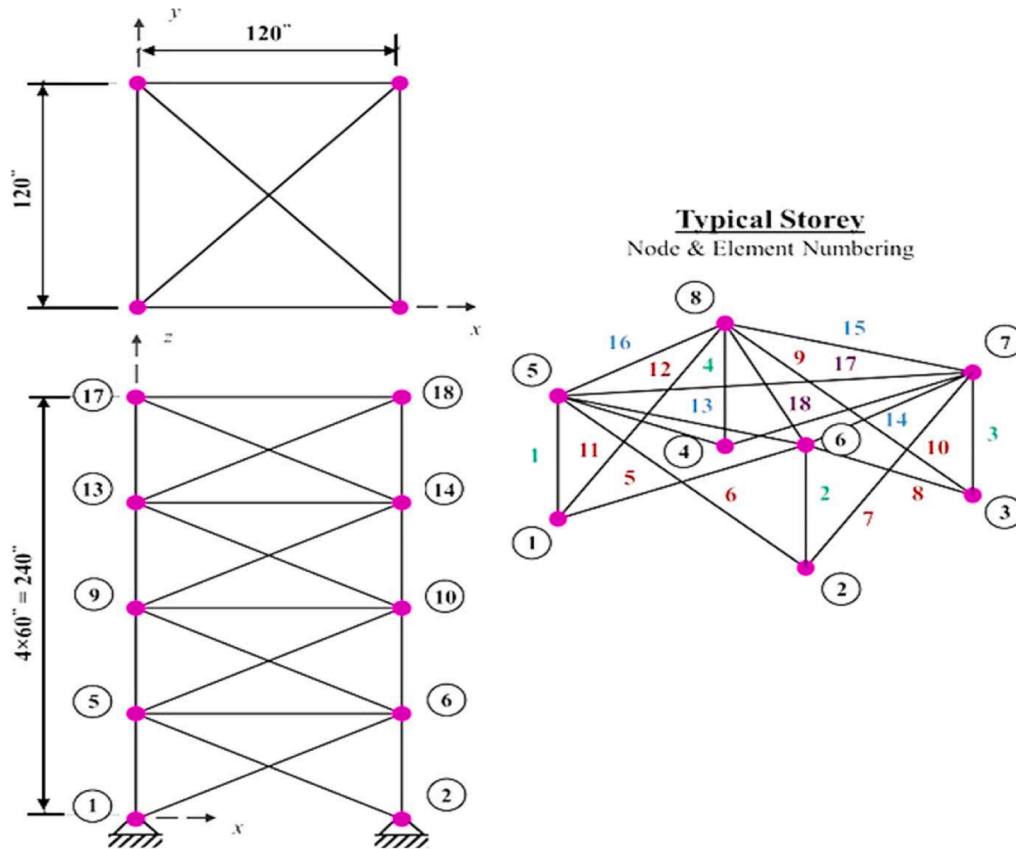


Fig. 6. 72 bar tower truss.

discontinuous Pareto fronts. Achieving optimal solution sets with MOLCA takes numerous function evaluations that produce significant increases in computational load. MOWAA implements an information feedback system which enables real-time adjustments of population movement to address these challenges. Through its adaptive mechanism MOWAA automatically optimizes its performance for different optimization problem landscapes which produces faster convergence and handles reduced operational costs. MOWAA maintains optimal convergence-diversity equilibrium through its function evaluation process which results in superior computational performance than MOLCA.

When applied to highly constrained optimization problems MOEDO demonstrates both poor distribution of Pareto front solutions and slow convergence speed. The integration of dominance-based sorting together with local search operators creates excessive exploitation which limits its ability to explore different areas of the search space. The computational complexity of MOEDO becomes increasingly difficult to manage when the optimization problem incorporates numerous objectives and constraints which reduces its effectiveness for large-scale truss optimization applications. MOWAA addresses these limitations through its dynamic search field system which combines personal and global best positions with weighted values. The approach distributes solutions consistently throughout the objective space to stop early convergence yet maintain diverse solutions. The structured exploration mechanism of MOWAA which combines Levy flight and boundary-driven exploration allows the algorithm to efficiently handle complex constraint landscapes.

The metaheuristic algorithm MORIME alongside its counterparts aims to discover optimal Pareto front solutions yet faces difficulties sustaining equilibrium between solution convergence and diversity. The selection process of the algorithm sometimes results in stagnation because it stops producing new non-dominated solutions after reaching a specific iteration limit. MORIME demonstrates poor performance in

high-dimensional optimization scenarios because it fails to establish an adaptive mechanism for balancing global exploration against local exploitation. MOWAA implements a dual-phase search system that dynamically switches between exploration and exploitation phases according to real-time performance feedback. The approach prevents solution stagnation while effectively investigating the search space to locate high-quality non-dominated solutions. MOWAA implements adaptive weight control to balance exploration and convergence while achieving superior results compared to MORIME in extensive truss design applications.

The current state-of-the-art algorithms demonstrate clear limitations which demand a new method that achieves effective balance between convergence and diversity. MOWAA fills these gaps by implementing weighted average position strategy and dynamic search adaptation with information feedback mechanism. MOWAA surpasses NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME by delivering a well-spread Pareto front while avoiding premature convergence and achieving efficient search space exploration for multi-objective truss optimization problems. The improved features of MOWAA establish it as an advanced solution that optimizes complex challenges with enhanced precision and operational speed and computational processing abilities.

The comparative performance analysis of multi-objective optimization algorithms-NSGA-II, MOEA/D, MOLCA, MOEDO, MORIME and MOWAA was done on eight benchmark truss structures, with consideration given to two main objectives, weight minimization and minimization of compliance (structural performance maximization). The best-obtained Pareto fronts for all the algorithms are shown in Figs. 10–17.

Among the many algorithms, MOWAA spanned the entire objective space encompassing all solutions, which distributed the Pareto fronts, allowing a higher density of solutions. MOWAA always produced higher HV values as it covered more objective space to reach high-quality solutions. This indicates that the coverage obtained from MOWAA Pareto

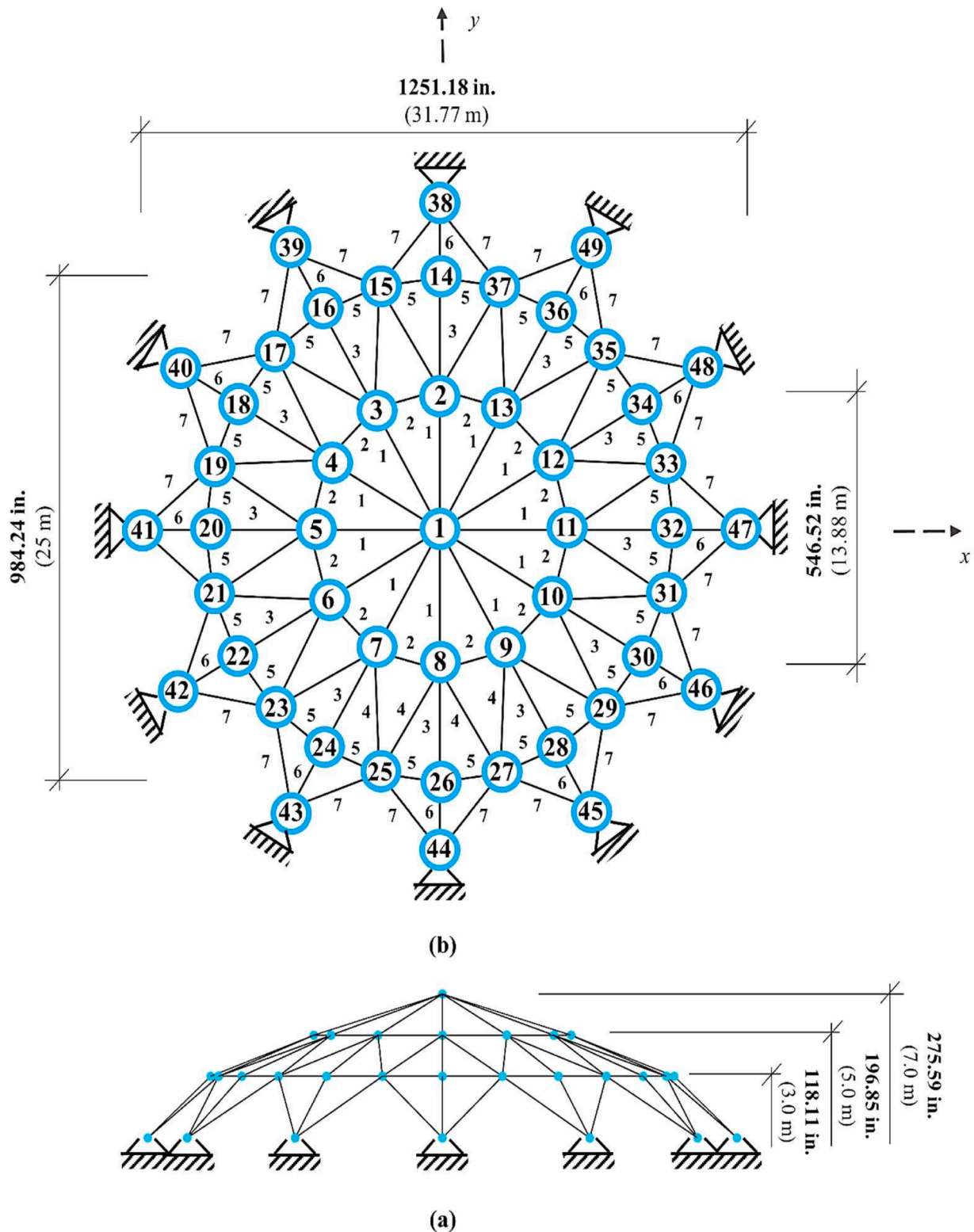


Fig. 7. 120 bar Truss Structure.

fronts furnishes a finer and more diverse area of non-dominated solutions than the other multi-objective algorithms.

From the results, it is established that MOWAA navigated and systematically explored the design space in a far superior manner than any of the other algorithms tested. Higher HV values for MOWAA indicate it not only approached True Pareto Front efficiently but also had a well-balanced spread across the objective space. This is due to MOWAA

effectively executed balance between convergence and diversity, crucial in deriving strong multi-objective solutions in truss design. Greater diversity in the Pareto fronts achieved in MOWAA implies that MOWAA effectively sought trade-offs between weight minimization and minimization of compliance because of its method of adaptively spreading solutions. Compared to NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME, MOWAA consequently returns non-dominated solutions

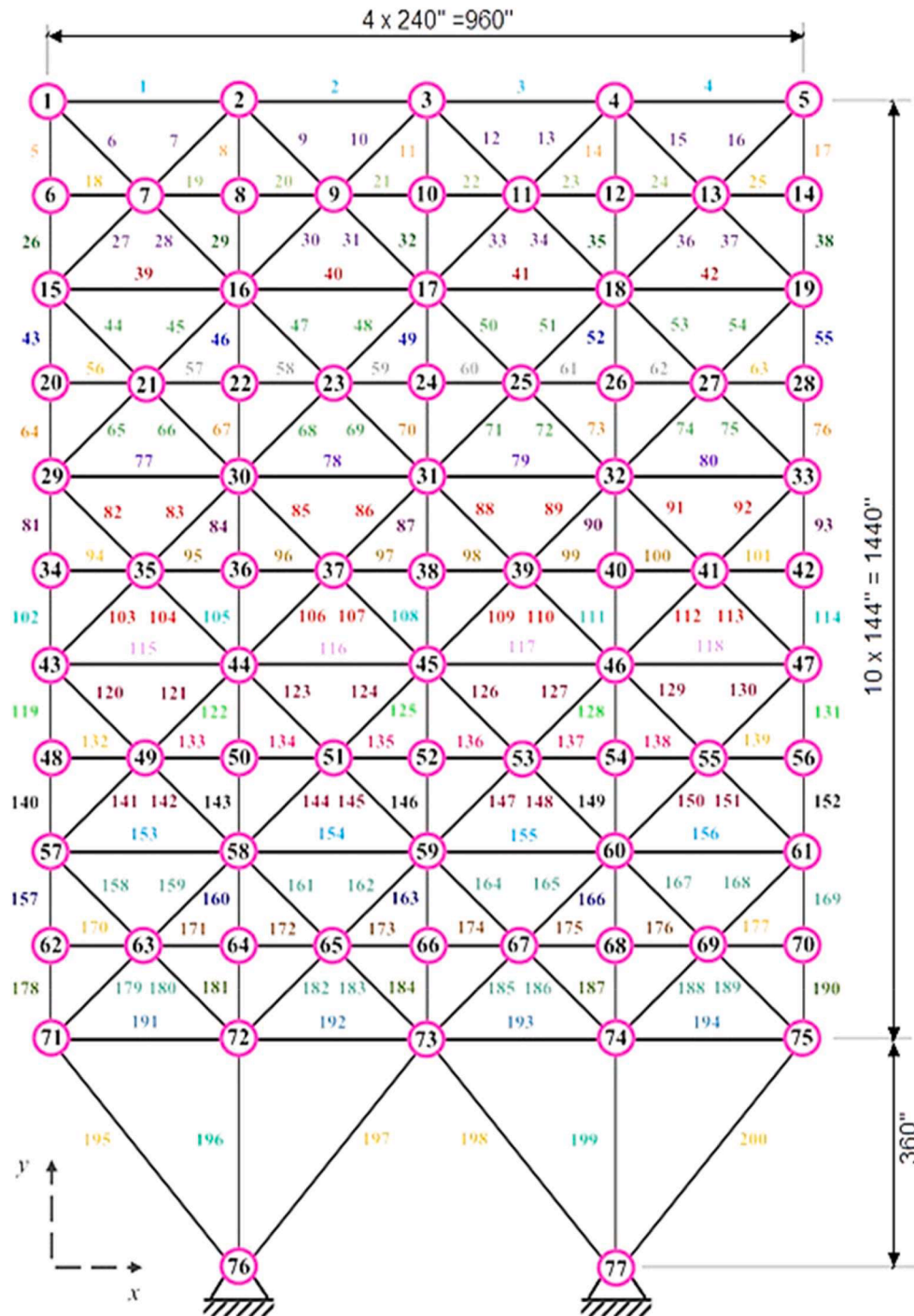


Fig. 8. 200 bar Truss.

spread over a much larger extent of objective space, thus boasting superior convergence and diversity. These qualities are most significant in multi-objective truss optimization.

Figs. 10–17 present the Pareto optimality achieved by different algorithms NSGA-II, MOEA/D, MOLCA, MOEDO, MORIME and MOWAA across various truss configurations, ranging from simpler 10-bar and 25-bar structures to more complex 200-bar and 942-bar structures. For simpler configurations, multiple algorithms yielded closely clustered Pareto fronts, indicating effective optimization with minimal trade-offs between objectives like weight and performance. As truss complexity increased, the spread of the Pareto fronts widened, highlighting the increased difficulty in balancing conflicting objectives, such as weight

minimization versus minimization of compliance, across a more intricate optimization landscape.

HV was used to quantify both convergence and diversity among the algorithms, with smaller truss configurations showing high HV values with relatively low function evaluation counts, indicating effective optimization with minimal computational cost. However, as truss complexity increased, the computational demand rose considerably, with evaluation counts reaching exceptionally high numbers, particularly for the 942-bar structure.

The results illustrate that maintaining convergence and diversity across objectives becomes increasingly challenging as truss complexity escalates. For simpler trusses, multiple algorithms provided efficient,

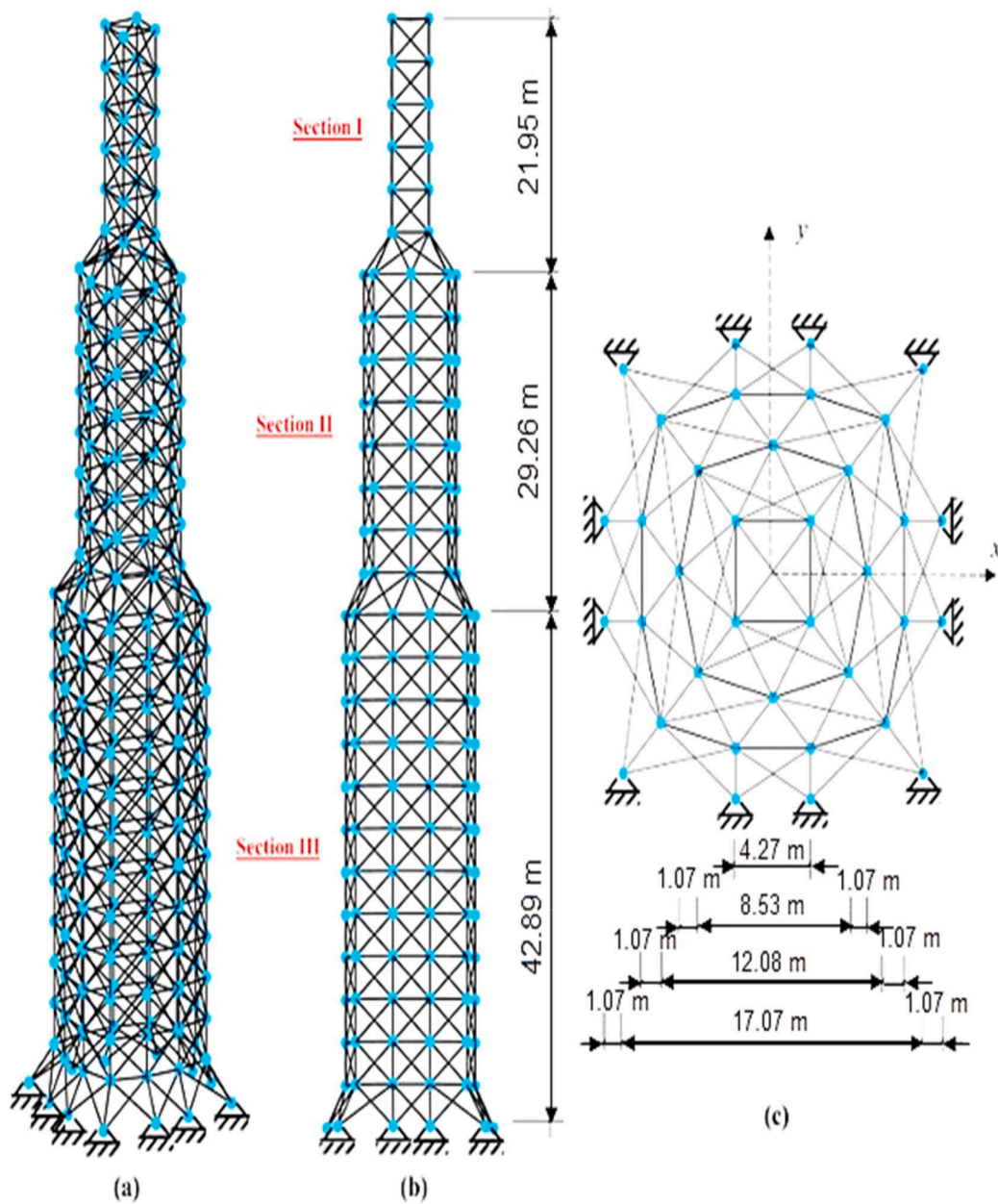


Fig. 9. 942 bar tower truss.

well-clustered Pareto fronts. However, with complex configurations, the ability to maintain a well-spread Pareto front diminished for many algorithms due to the added difficulty of navigating conflicting objectives and a more rugged optimization landscape.

MOWAA consistently achieved a more converged and distributed Pareto front across all configurations, especially in complex structures, compared to other algorithms. This superior performance highlights MOWAA capability in handling high-dimensional, multi-objective truss optimization problems by providing both a wider range of trade-offs and higher convergence efficiency.

4.2. Convergence analysis through HV metric

Let $z^r = (Z_1^r, \dots, Z_m^r)^T$, be a worst point dominated by all the Pareto optimal objective vectors. The HV of P is defined as the volume of the objective space dominated by solutions in P and bounded by z^r .

$$HV(P) = \text{VOL} \left(\bigcup_{z \in P} [z_1, Z_1^r] \times \dots \times [z_m, Z_m^r] \right) \quad (15)$$

where VOL indicates the Lebesgue measure. Hyper Volume assesses the portion of the objective space occupied by the non-dominated solution set; higher values indicate better performance as per Eq. (15) [22].

Fig. 18 shows the HV parameters of NSGA-II, MOEA/D, MOLCA, MOEDO, MORIME and MOWAA for a typical truss problem. This shows the ability and strength of MOWAA to produce high quality Pareto fronts. Compared to, algorithms such as NSGA-II and MORIME, MOWAA show significant improvement over HV, especially for complex problems where computational efficiency and quality of solutions are important. The comparison in these boxes shows that MOWAA is better than other algorithms, achieving the highest average HV and the lowest IQR. The small density in the centre of the boxplots of MOWAA ensures that it can pass through the selected area, find a good solution and do this without excessive monitoring. In general, MOWAA exhibits better performance in system optimization, especially for complex multi-objective problems

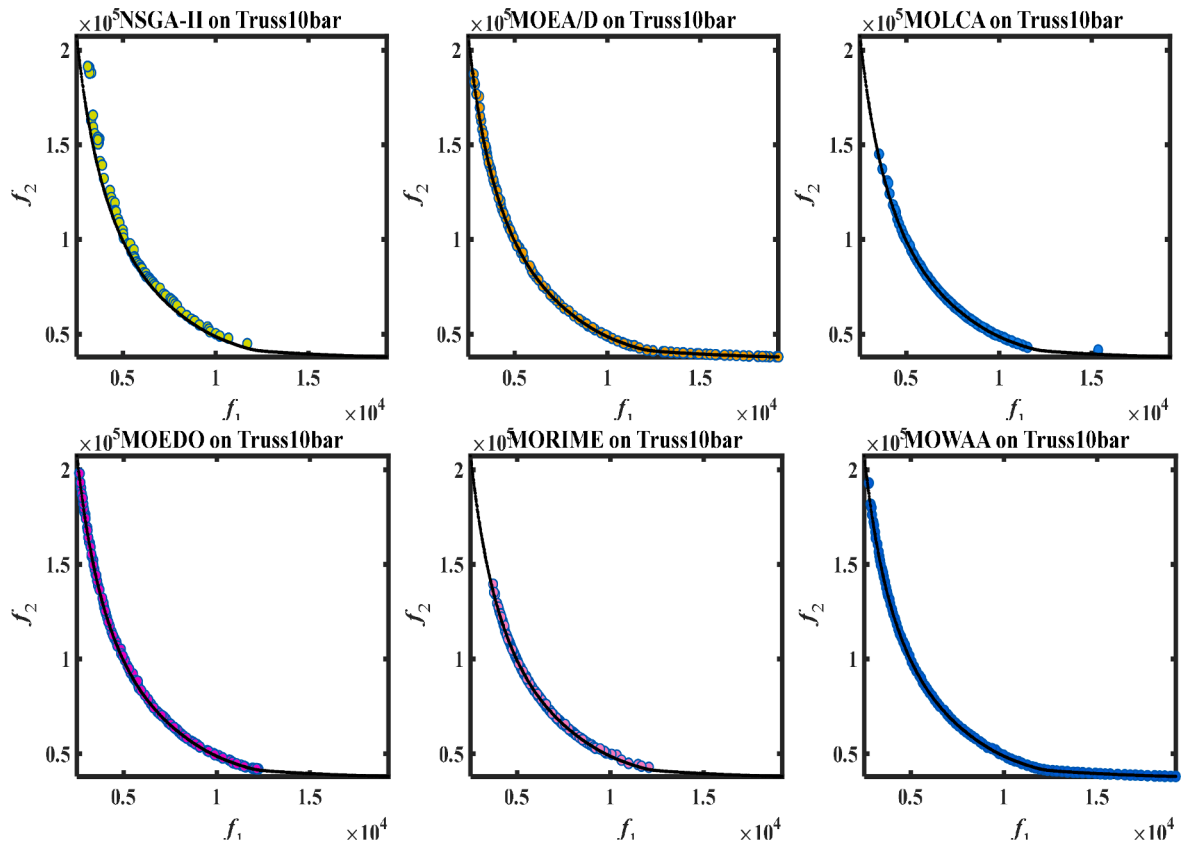


Fig. 10. Best pareto fronts for 10-bar Truss using different algorithms.

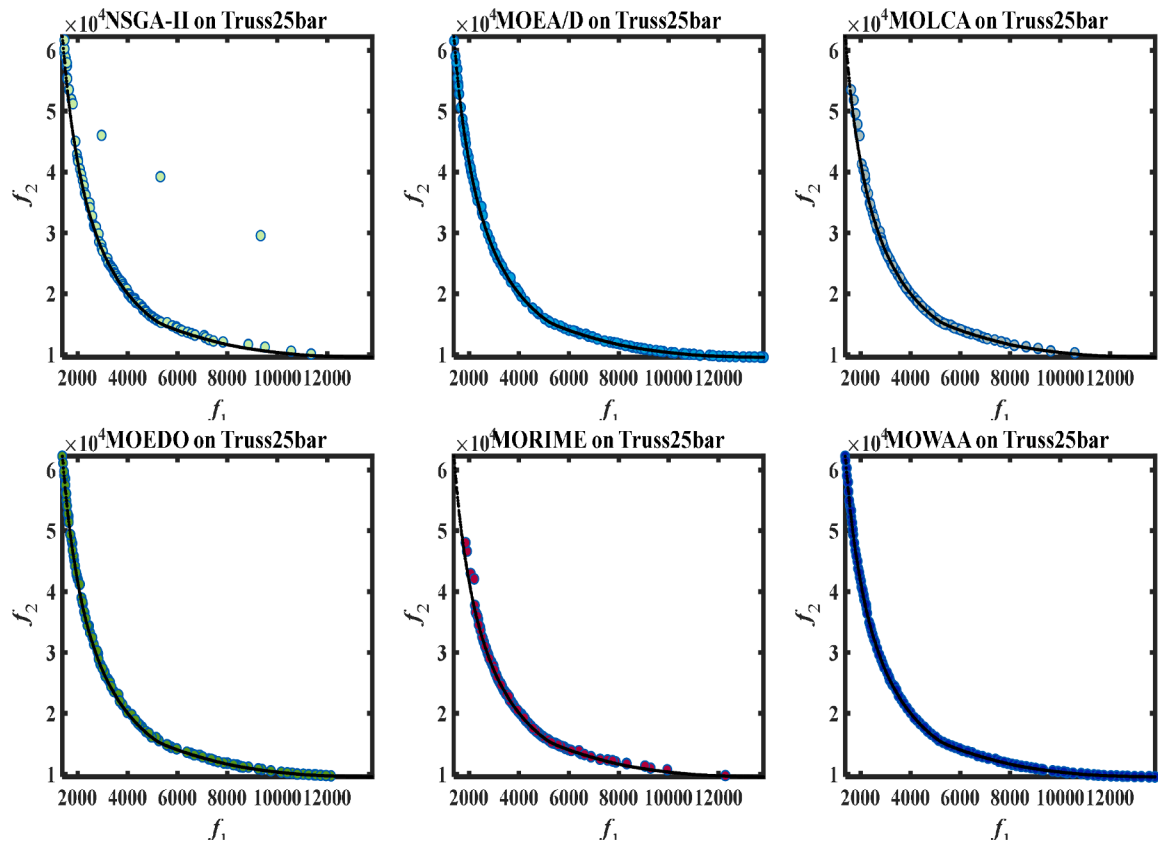


Fig. 11. Best pareto fronts for 25-bar Truss using different algorithms.

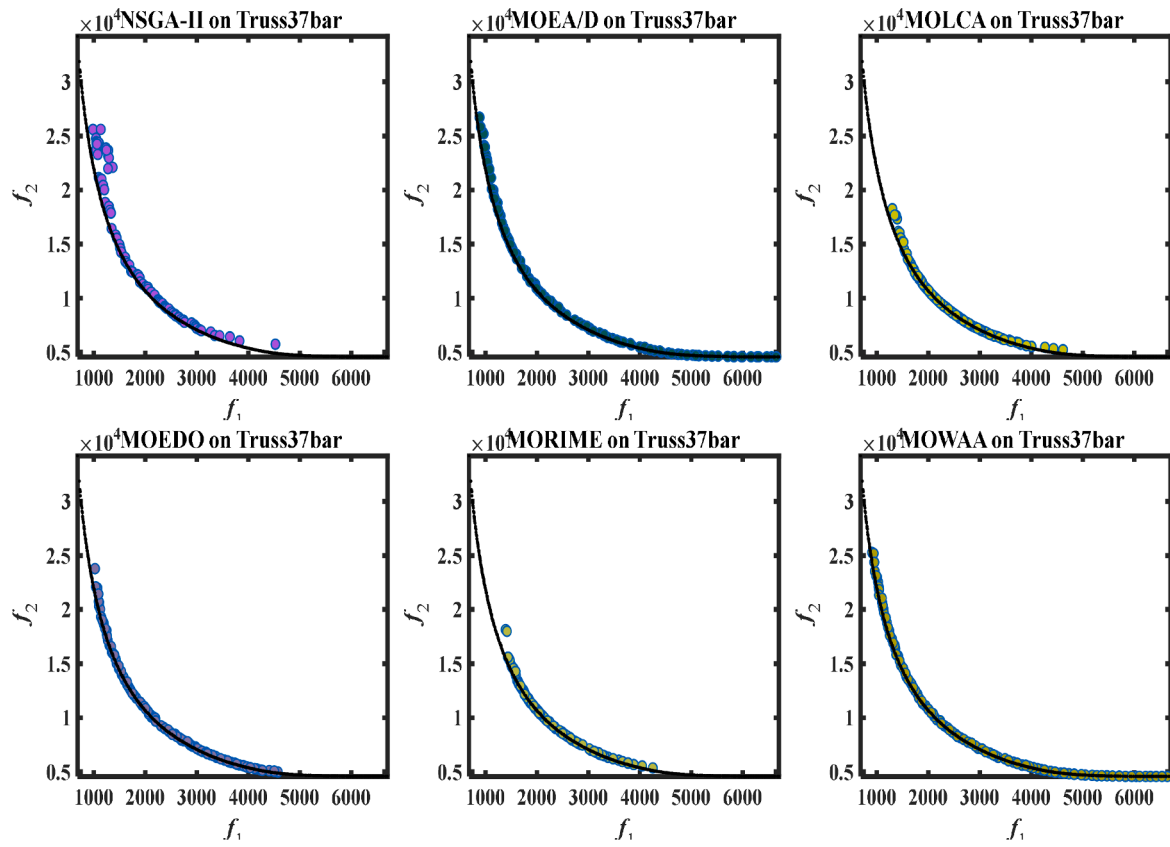


Fig. 12. Best pareto fronts for 37-bar Truss using different algorithms.

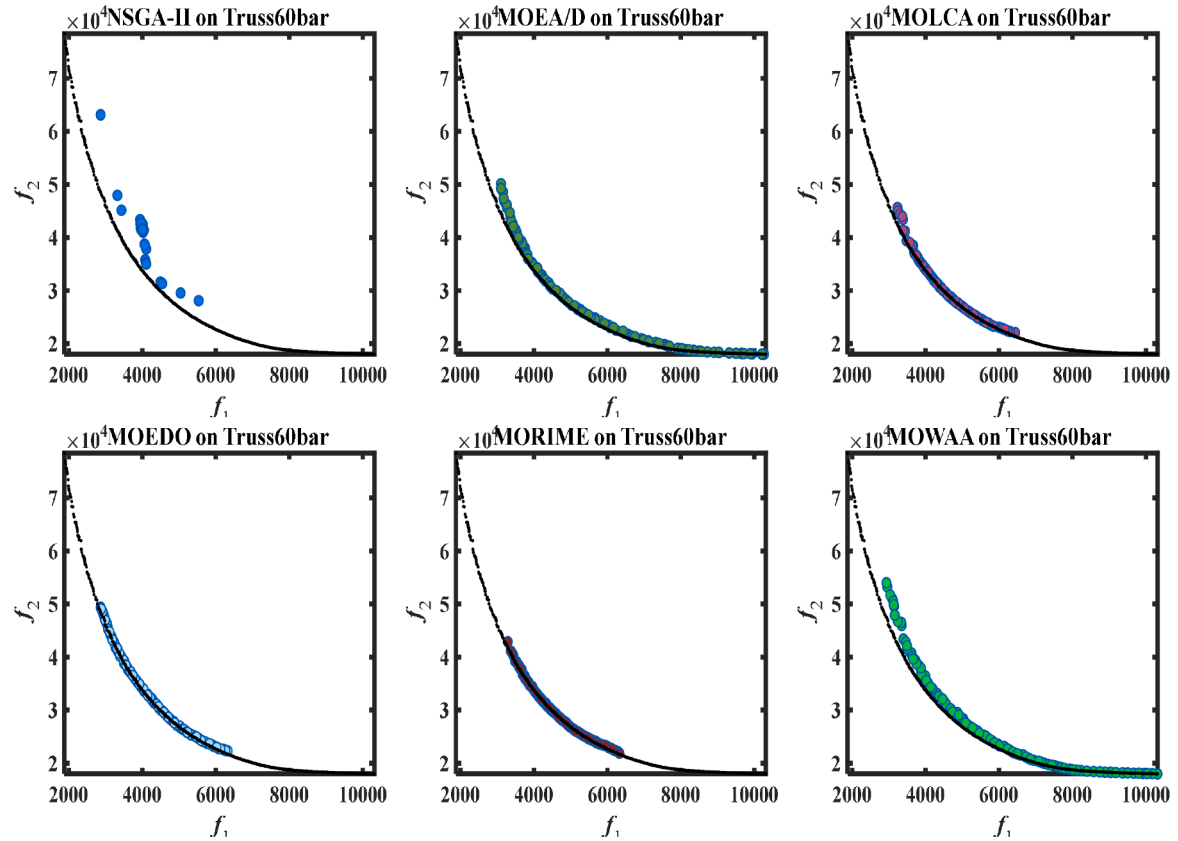


Fig. 13. Best pareto fronts for 60-bar Truss using different algorithms.

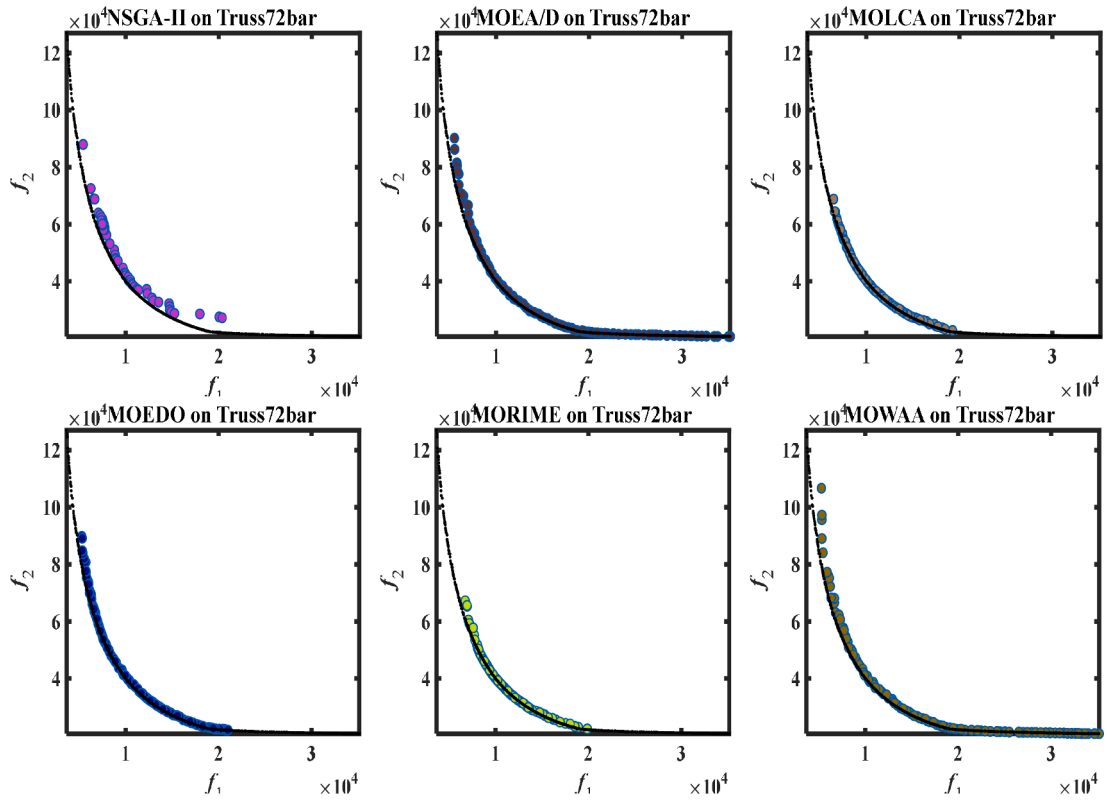


Fig. 14. Best pareto fronts for 72-bar Truss using different algorithms.

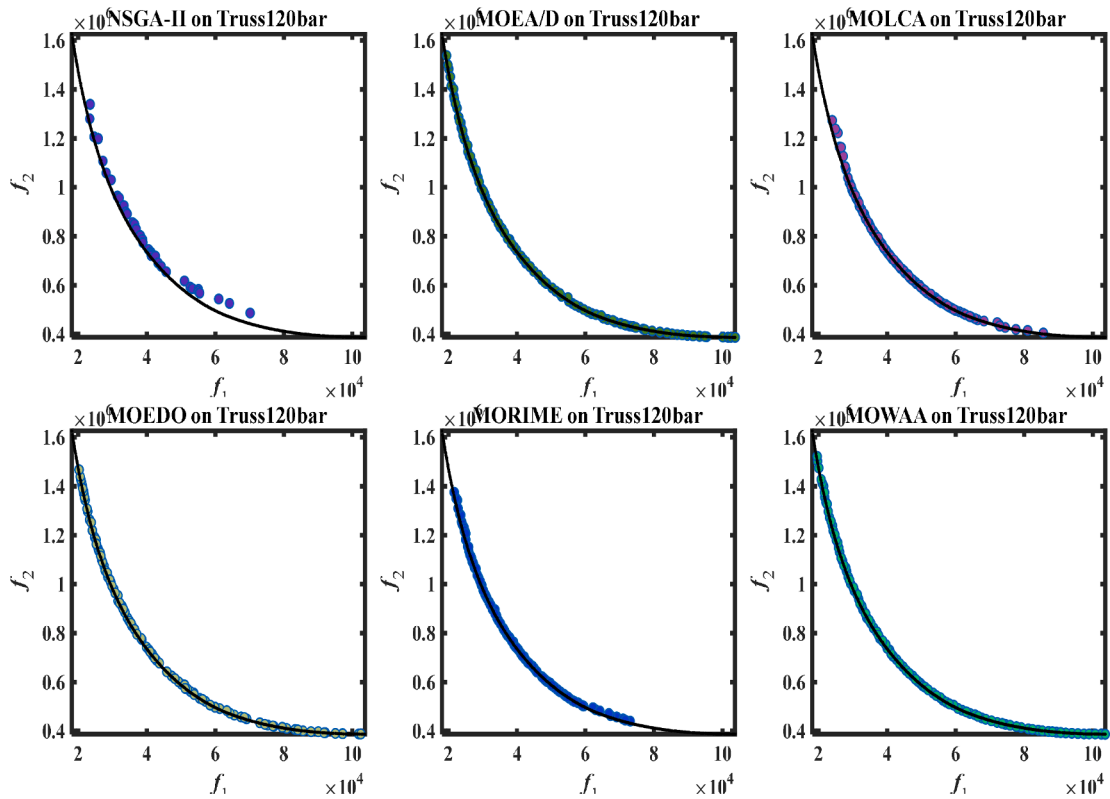


Fig. 15. Best pareto fronts for 120-bar Truss using different algorithms.

that require robust, diverse and well-organized solutions.

Fig. 19 shows the cross-sectional HV information of over eight benchmark constraint-based design problems, comparing the

performance of MOWAA with several commonly used evolutionary algorithms (MOEA): NSGA-II, MOEA/D, MOEDO, MORIME and MOLCA. These regressions show that MOWAA performs better than other

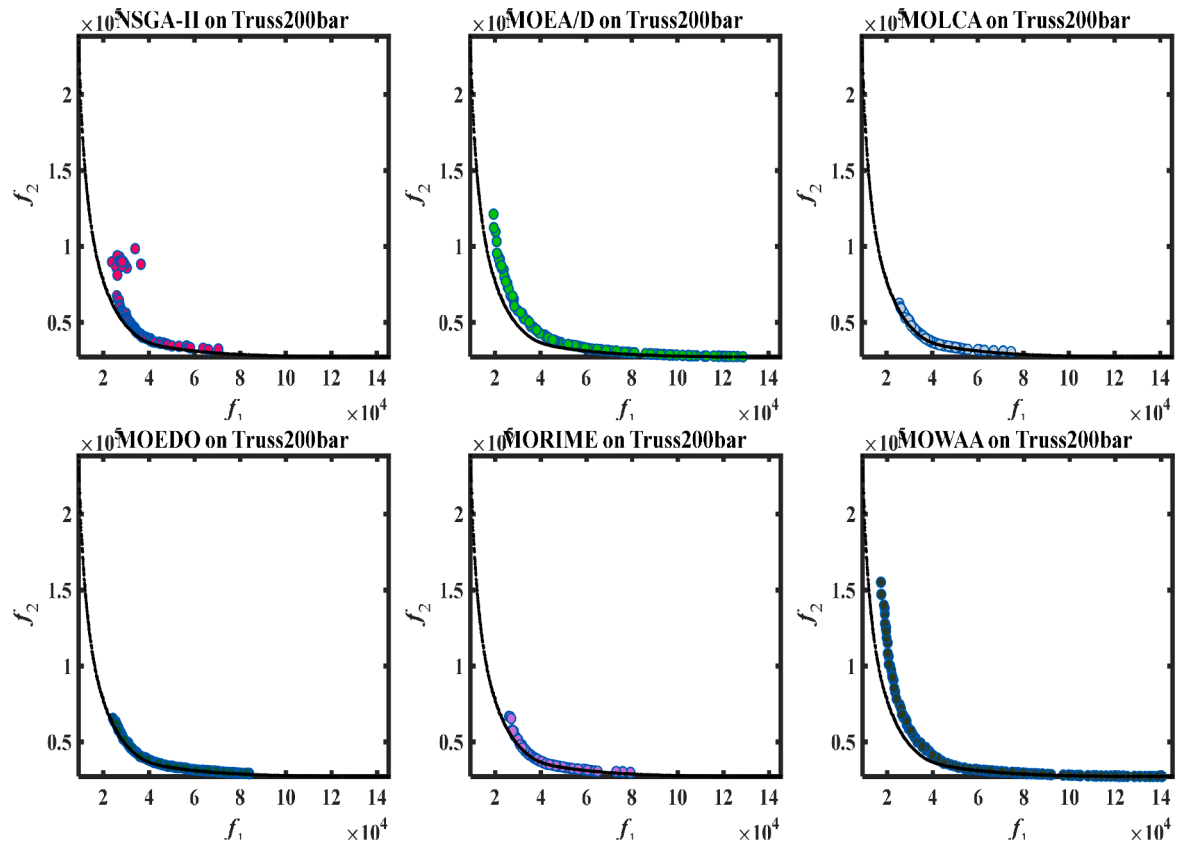


Fig. 16. Best pareto fronts for 200-bar Truss using different algorithms.

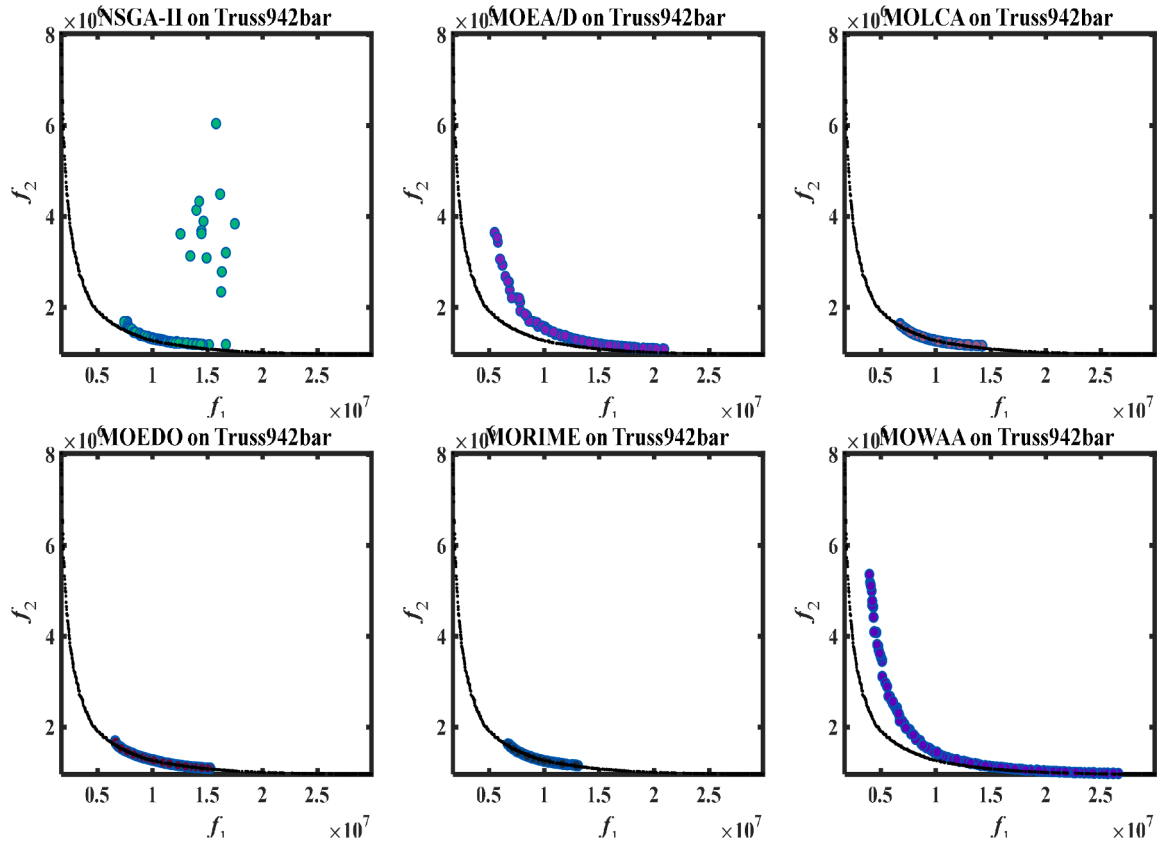


Fig. 17. Best pareto fronts for 942-bar Truss using different algorithms.

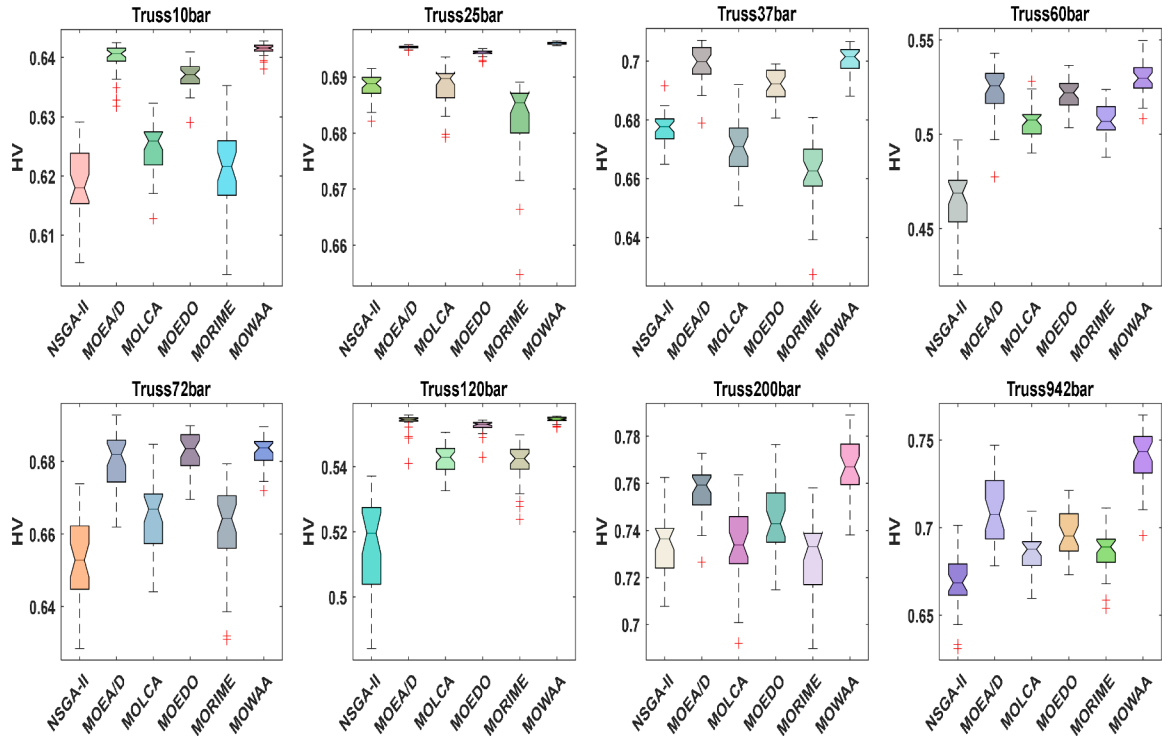


Fig. 18. Box plot of hyper volume by considered algorithms for different truss structures.

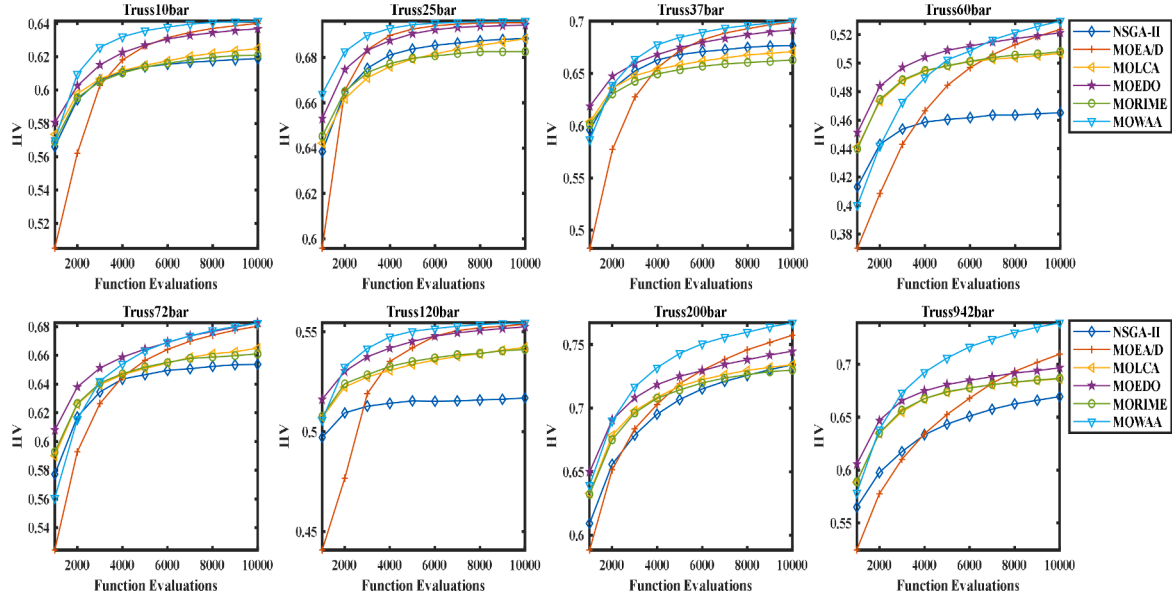


Fig. 19. Convergence plot of hyper volume by considered algorithms for different truss structures.

algorithms, as shown by faster convergence and higher final HV values than all benchmark problems. MOWAA convergence curve at the beginning shows a significant increase in the HV value, which shows its ability to quickly find a superior solution and approach the Pareto optimal distribution. This first improvement in HV exceeds the improvement of other algorithms, increasing MOWAA capabilities in terms of solution quality and diversity. The design of the MOWAA algorithm allows it to cover a large part of the solution area quickly and ensure a good trade between the variables without sacrificing accuracy. Fast exploration of the objective space is critical for multi-objective optimization, as it makes navigating through complex spaces more efficient.

In comparison to the algorithms under consideration, MOWAA shows a consistently higher final HV, indicating that it can regulate convergence and diversity with a smaller number of iterations. These results demonstrate the ability of MOWAA to quickly obtain optimal solutions, this is a great advantage for multi-objective optimization problems related to mechanical design. Unlike other algorithms, which show rapid changes in HV values, MOWAA continuous improvement in HV shows its ability to find different solutions in the solution space. It is evident that, the conversion curves support the validity of MOWAA for complex design optimization problems, as it achieves faster convergence and higher final HV values compared to NSGA-II, MOEA/D, MOEDO, MORIME and MOLCA.

The HV values in Table 1 show the performance of MOWAA across different constraint classification problems, comparing its performance with other multi-objective optimization algorithms under the difficulty of the classification. For the simple 10-step algorithm, MOWAA obtained a HV of 0.64143, a value higher than that reached by other algorithms, showing its first step in generating a well-distributed solution set. This trend continues with the 25-bar truss, where MOWAA achieved a HV of 0.69606, in line with its peers.

As complexity rises with the 37-bar truss, MOWAA maintains its advantage, achieving an HV of 0.70011. This consistent performance signals MOWAA ability to cover the objective space effectively while maintaining high solution quality. For the 60-bar truss, MOWAA demonstrated robust performance with an HV of 0.52943, indicating its adaptability to handle larger structural design challenges without compromising on efficiency.

In case of a 72-bar truss, MOWAA achieved a HV of 0.68247, continuing the trend of convergence and diversity compared to other algorithms. This principle was demonstrated in a 120-bar truss, where MOWAA reached 0.55442 HV, which confirmed its ability to provide a good solution for increasingly complex truss structures. The performance of MOWAA becomes more accurate on large optimization problems. For the 200-bar truss, MOWAA achieved an impressive 0.76701 HV, demonstrating the ability to clearly travel through extended spans. This advantage is confirmed again in the complex problem of the 942-bar truss, where MOWAA achieves 0.73911 HV and outperforms all other algorithms. This performance demonstrates MOWAA ability to provide efficient optimization solutions in highly complex environments.

Table 2 demonstrates that MOWAA consistently ranks as the top-performing algorithm across a series of benchmark truss optimization problems, with a mean rank of 5.7143 for the 10-bar truss, significantly outperforming other algorithms, with MOEA/D being its closest rival. MOWAA continues to hold the highest ranking with a score of 5.9714 in the 25-bar truss optimization and maintains a robust position with a rank of 5.4286 for the 37-bar truss problem. For the 60-bar truss, MOWAA demonstrates its efficiency with a rank of 5.3429, showing better performance than both MOEA/D and MOEDO. This pattern continues in the 72-bar truss problem, where MOWAA achieves a rank of 5.0286, reinforcing its superior performance over MORIME.

In the 120-bar truss scenario, MOWAA rank of 5.4571 highlights its advantage over MOEA/D, proving its effectiveness in generating high-quality solutions even for more complex structures. This trend persists with the 200-bar truss problem, where MOWAA reaches a rank of 5.6286, again surpassing MOEA/D. The highly complex 942-bar truss problem further emphasizes MOWAA capability, where it attains the top rank of 5.6857, demonstrating superior performance as structural complexity intensifies.

This comparative analysis performance of MOWAA with five optimization methods - NSGA-II, MOEA/D, MORIME, MOLCA and MOEDO - evaluates performance based on the average ranking for eight benchmark optimization problems. The ranking is supported by statistical analysis, with p -values ranging from 1.81E-32 to 6.99E-19, confirming the importance of MOWAA work. The high p -value supports MOWAA superiority over all complexities as it outperforms the other considered

algorithms. Although algorithms such as MOEA/D and MORIME successfully complete the challenge in some cases, they do not match the performance and stability of MOWAA on all problems.

4.3. Diversity analysis by IGD metric

Given P^* as a set of points uniformly sampled along the PF and P as the set of solutions obtained from an EMO algorithm. The IGD value of P is calculated as per Eq. (16).

$$IGD(P, P^*) = \frac{\sum_{z \in P^*} \text{dist}(z, P)}{|P^*|} \quad (16)$$

where $\text{dist}(z; P)$ is the Euclidean distance between z and its nearest neighbour in P . To calculate the IGD, we need to sample enough points from the PF to form P^* . IGD evaluates the convergence of the obtained Pareto front to the True Pareto front by calculating the average distance between points on the produced front and their nearest counterparts on the actual front; lower values indicate better convergence [23].

Fig. 20 shows the values of the inversion distance (IGD) produced by various optimization algorithms (e.g., NSGA-II, MOEA/D, MOLCA, MORIME, MOEDO and MOWAA) in order of benchmark truss optimization problems. Box plots shown for various truss configurations including Truss10 bar, Truss25 bar, Truss37 bar, Truss60 bar, Truss72 bar, Truss120 bar, Truss200 bar and Truss942 bar illustrate the performance of this algorithm.

The IGD measurement is important to determine how effective the obtained solution is in approximating the true Pareto front; A low IGD value indicates that the solution is close to optimal. In the Truss10 bar optimization problem, MOWAA shows a lower median IGD compared to similar problems, indicating the robustness of convergence in this case. The MOWAA boxplot shows the narrowest interquartile range (IQR), indicating consistent performance with little variability across different benchmark problems. On the other hand, the results of the MOLCA and MOEDO algorithms show a broad IGD distribution, indicating the difficulty in achieving convergence like the Pareto front. In addition, the performance of MOEA/D, MORIME and NSGA-II is characterized by high IGD values and wide spreads, which enhances their performance in reliably finding optimal solutions. The importance of MOWAA is confirmed when results of the Truss25 bar problem are analysed, because it records the lowest median IGD and the narrowest IQR, indicating that it produces stable and repeatable results algorithm. The IGD values associated with MOLCA and MOEDO still show a significant difference, as shown by their wide box and outliers. These outliers present significant challenges for these algorithms to produce reliable solutions. In comparison, MOWAA low IQR and low IQR demonstrate its ability and reliability to approach the Pareto front even with more complex framework optimization problems. For the Truss37 bar configuration, MOWAA continues to demonstrate its excellence in IGD operations. The algorithm is expert in handling structural optimization problems of moderate complexity, demonstrating its robustness in solution of complex structural optimization problems. In the Truss60 bar structure, MOWAA also achieves a medium IGD and a lower IQR, thus increasing its reliability in more complex optimization tasks. It is

Table 1
Hypervolume values for different Truss structures using considered algorithms.

Problem	M	D	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA
Truss10bar	2	10	6.1880e-1 ± 6.28e-3	6.3995e-1 ± 2.61e-3	6.2495e-1 ± 4.38e-3	6.3683e-1 ± 2.26e-3	6.2087e-1 ± 7.54e-3	6.4143e-1 ± 9.93e-4
Truss25bar	2	8	6.8838e-1 ± 2.07e-3	6.9537e-1 ± 2.64e-4	6.8832e-1 ± 3.54e-3	6.9435e-1 ± 4.94e-4	6.8263e-1 ± 7.15e-3	6.9606e-1 ± 2.10e-4
Truss37bar	2	15	6.7673e-1 ± 5.60e-3	6.9921e-1 ± 6.23e-3	6.7079e-1 ± 9.73e-3	6.9199e-1 ± 5.23e-3	6.6286e-1 ± 1.18e-2	7.0011e-1 ± 4.62e-3
Truss60bar	2	25	4.6509e-1 ± 1.63e-2	5.2361e-1 ± 1.37e-2	5.0682e-1 ± 9.05e-3	5.2122e-1 ± 8.31e-3	5.0805e-1 ± 8.67e-3	5.2943e-1 ± 8.49e-3
Truss72bar	2	16	6.5372e-1 ± 1.14e-2	6.8030e-1 ± 7.59e-3	6.6497e-1 ± 9.92e-3	6.8264e-1 ± 5.52e-3	6.6108e-1 ± 1.33e-2	6.8247e-1 ± 4.11e-3
Truss120bar	2	7	5.1685e-1 ± 1.37e-2	5.5378e-1 ± 2.68e-3	5.4211e-1 ± 4.29e-3	5.5243e-1 ± 2.02e-3	5.4114e-1 ± 6.07e-3	5.5442e-1 ± 8.11e-4
Truss200bar	2	29	7.3397e-1 ± 1.31e-2	7.5728e-1 ± 1.06e-2	7.3398e-1 ± 1.65e-2	7.4469e-1 ± 1.45e-2	7.2995e-1 ± 1.53e-2	7.6701e-1 ± 1.17e-2
Truss942bar	2	59	6.6939e-1 ± 1.68e-2	7.0973e-1 ± 1.93e-2	6.8616e-1 ± 1.15e-2	6.9676e-1 ± 1.33e-2	6.8679e-1 ± 1.24e-2	7.3911e-1 ± 1.60e-2

Table 2

p-values and rank based on hyper volume for different truss structures.

Problem	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA	<i>p</i> -value
Truss10bar	1.6571	5.2	2.5429	4.0571	1.8286	5.7143	4.78E-31
Truss25bar	2.2857	5	2.2857	4.0286	1.4286	5.9714	1.81E-32
Truss37bar	2.6571	5.2571	2.0286	4.2571	1.3714	5.4286	3.80E-30
Truss60bar	1	4.6571	2.6857	4.3714	2.9429	5.3429	9.44E-26
Truss72bar	1.5143	4.6571	2.6571	4.9714	2.1714	5.0286	8.86E-25
Truss120bar	1.0857	5.2571	2.5429	4.2	2.4571	5.4571	4.11E-31
Truss200bar	2.5714	4.6	2.5714	3.5429	2.0857	5.6286	6.99E-19
Truss942bar	1.5143	4.5429	2.7714	3.7143	2.7714	5.6857	6.18E-22

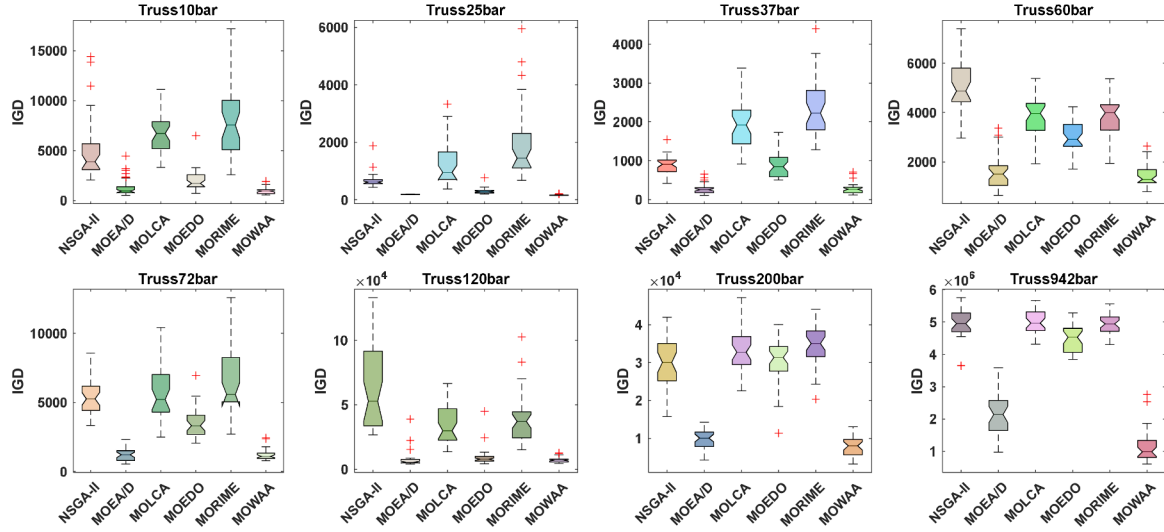


Fig. 20. Box plot of IGD values by considered algorithms for different truss structures.

expected that as the complexity of the problem increases (due to the large design space and additional constraints imposed), the median IGD for all algorithms increases. This feature is particularly evident in the Truss72 bar configuration products, where the MOWAA maintains a low mid-range IGD and low dispersion, increasing the ability to produce consistent solutions even as complexity increases. of the problem. On the Truss120 bar structure, complexity get tougher, MOWAA reaches the

competitive IGD median, but the IQR is growing a little. This increase in variance is a common feature observed among all algorithms when dealing with more complex configurations. In the case of Truss200 bar, MOWAA still shows good performance, reaching a low median IGD even though the IQR increases slightly. These observations show that MOWAA can still provide solutions close to the Pareto front while generating fewer backends, thus confirming its ability to handle

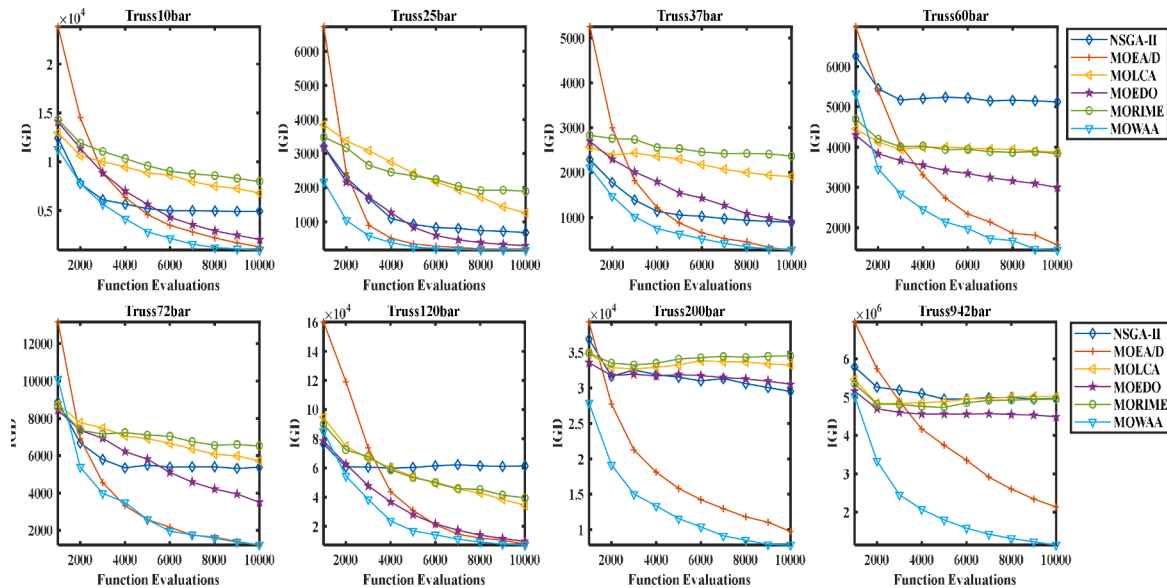


Fig. 21. Convergence plot of IGD by considered algorithms for different truss structures.

complex chassis optimization tasks. and size. Among the reference problems considered, the Truss942 bar problem is the most complex problem. In this highly complex problem, MOWAA still maintains a low median IGD value, although the IQR is marginally extended. In comparison, other algorithms achieve higher IGD values as well as wider distributions and higher flows of individuals, indicating that it is very difficult to achieve convergence. Even in the face of complex optimizations, MOWAA performance shows a remarkable level of robustness and adaptability, allowing it to be useful even if its variance is only slightly increased.

Fig. 21 provides correlation plots showing IGD metrics for various multi-objective optimization algorithms, including NSGA-II, MOEA/D, MOLCA, MORIME, MOEDO and MOWAA, on eight benchmark truss structure problems, from 10 bar truss to 942 -bar truss. The x-axis represents the number of performance assessments performed and the y-axis represents the IGD value. In general, a lower IGD value indicates closer to the true Pareto front. These IGD plots show not only the efficiency of the algorithms but also their convergence rates, thus providing important insights into the performance aspects of the optimization process.

Table 3 provides a detailed evaluation of the performance of several benchmark truss problems of a multi-objective optimization using the inverse generating distance (IGD) metric as a measure of efficiency. In simple optimization tasks, such as the 10-bar lattice design problem, MOWAA obtained an IGD value of $9.6226e+2$. These results show that MOWAA can handle simple truss structure problems efficiently, making it a suitable for such problems. As the complexity of the problem increases, for example in the case of the 25-bar lattice design, the IGD written by MOWAA improves to $1.6657e+2$. This finding demonstrates the flexibility of the algorithm and highlights its ability to maintain strong performance even for a difficult problem, demonstrating its advantages in various optimization tasks. When solving moderate structural problems such as 60 bar and 72 bar deformations, MOWAA reports an IGD of $2.7682e+2$ and $1.4415e+3$, respectively. These figures highlight the robustness of the algorithm and its ability to approximate the Pareto optimal set, even in difficult optimization environments. The robustness and flexibility of MOWAA is particularly evident when applied to the 942-bar truss problem, where it achieves an IGD of $1.1430e+6$. This state-of-the-art optimization demonstrates the algorithm ability to handle complex constraints while providing competitive performance regardless of problem size and complexity. Although MOWAA does not always produce the lowest IGD metrics for all tests, its efficiency and closeness to optimal solutions make it promising for difficult optimization tasks. The algorithms perform well in both low- and high-order problems, producing reliable results close to the exact Pareto front, while performing well in design optimization. The ability of MOWAA to remain competitive with IGD when faced with major challenges, such as the 942-bar truss problem, demonstrates its ability to improve engineering performance. Although MOWAA does not achieve the lowest IGD for all problems, it is competitive and reliable.

Table 4 provides a detailed comparison of p -values for IGD comparisons of MOWAA with several leading multi-objective optimization algorithms on eight different truss optimization problems. The p -value indicates that the performance of MOWAA is statistically significant,

making it effective under optimal conditions. For example, in the 10-bar vertical truss problem, MOWAA obtained a significant p -value of $1.70E-28$, indicating its high significance. This behaviour is consistent with other complex problems as shown for 25 and 37 bar truss strains, where the p -values are $1.48E-32$ and $1.80E-31$, respectively, making MOWAA more competitive than peers. As the complexity of the truss structure increases, MOWAA exhibits increasing efficiency. For example, values of $P-1.53E-27$ and $2.75E-27$ show significant performance differences for problem of 60 bar and 72 bar. In the complex 942-bar truss optimization problem, MOWAA achieved a p -value of $5.65E-27$ and demonstrated its strength even under extreme conditions.

This consistent set of low p -values across different problems makes MOWAA a reliable choice for computational multiobjective optimization problems in truss design. In addition, the Friedman rank analysis in Table 4 provides an overview of the relative performance of MOWAA. Lower scores on the Friedman test indicate better outcomes. For the 10-bar truss problem, MOWAA scored 1.5143, demonstrating better performance in simple problems. As for the 60-pole truss, the MOWAA level of 1.5429 indicates that it continues to perform well. In addition, for the more complex 942-bar truss problem, MOWAA achieved a score of 1.1714, indicating its ability to cope well with complex optimization scenarios.

4.4. Quality assessment of non-dominated solutions by spacing matrix

Spacing measures the evenness of spacing between adjacent solutions along the Pareto front, aiming for a balanced distribution without clustering or excessive gaps. Lower SP values indicate more uniformly spaced solutions as per Eq. (17).

$$\text{Spacing} = \sqrt{\frac{1}{N_0 - 1} \sum_{i=1}^{N_0} \left(\text{mean}(d_i^{\text{spacing}}) - d_i^{\text{spacing}} \right)^2} \quad (17)$$

Where, N_0 denoted the number of pareto fronts and d_i^{spacing} denotes the Euclidean distances between consecutive solutions in the objective space. Spacing assess the spread and distribution of solutions across the Pareto front [24].

The box plots depicted in Fig. 22 illustrate the spacing values corresponding to various optimization challenges in truss structures, as evaluated by multiple multi-objective algorithms, including NSGA-II, MOEA/D, MOLCA, MORIME, MOEDO and MOWAA. The spacing metric is an essential indicator of the uniformity of solution distribution along the Pareto front; lower spacing values signify a more evenly distributed set of solutions, which is generally preferred in optimization scenarios. MOWAA stands out for its consistently superior performance across all tested truss structures, demonstrating an impressive capacity for both spacing distribution and convergence.

The boxplots show that MOWAA achieves a lower and more stable spacing values compared to other algorithms, showing more uniform solutions on the Pareto front. This capability of the algorithm is important to accurately determine the location of the solution, which ultimately increases the quality and reliability of the algorithm. MOWAA ability to maintain this uniformity among different truss systems demonstrates its usefulness in solving complex optimization

Table 3
IGD values for different Truss structures using considered algorithms ($M = 2$).

Problem	D	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA
Truss10bar	10	$4.9442e+3 \pm 3.03e+3$	$1.3080e+3 \pm 9.04e+2$	$6.7897e+3 \pm 2.04e+3$	$2.0418e+3 \pm 1.04e+3$	$7.9889e+3 \pm 3.71e+3$	$9.6226e+2 \pm 3.55e+2$
Truss25bar	8	$6.8255e+2 \pm 2.47e+2$	$1.8718e+2 \pm 8.37e+0$	$1.2560e+3 \pm 7.46e+2$	$2.9873e+2 \pm 1.02e+2$	$1.8959e+3 \pm 1.23e+3$	$1.6546e+2 \pm 1.43e+1$
Truss37bar	15	$8.8742e+2 \pm 2.13e+2$	$2.6839e+2 \pm 1.26e+2$	$1.9100e+3 \pm 6.09e+2$	$8.9447e+2 \pm 3.55e+2$	$2.3660e+3 \pm 7.10e+2$	$2.7682e+2 \pm 1.34e+2$
Truss60bar	25	$5.1187e+3 \pm 1.07e+3$	$1.5726e+3 \pm 6.68e+2$	$3.8753e+3 \pm 7.69e+2$	$2.9989e+3 \pm 6.78e+2$	$3.8363e+3 \pm 8.11e+2$	$1.4415e+3 \pm 4.86e+2$
Truss72bar	16	$5.3936e+3 \pm 1.33e+3$	$1.2189e+3 \pm 4.55e+2$	$5.7283e+3 \pm 2.02e+3$	$3.5128e+3 \pm 1.07e+3$	$6.5335e+3 \pm 2.67e+3$	$1.2256e+3 \pm 3.91e+2$
Truss120bar	7	$6.1365e+4 \pm 3.29e+4$	$7.7669e+3 \pm 6.39e+3$	$3.4276e+4 \pm 1.45e+4$	$9.6076e+3 \pm 7.14e+3$	$3.9334e+4 \pm 2.05e+4$	$7.0877e+3 \pm 2.08e+3$
Truss200bar	29	$2.9489e+4 \pm 6.99e+3$	$9.7517e+3 \pm 2.77e+3$	$3.3274e+4 \pm 5.99e+3$	$3.0525e+4 \pm 5.95e+3$	$3.4552e+4 \pm 5.83e+3$	$7.8339e+3 \pm 2.68e+3$
Truss942bar	59	$4.9652e+6 \pm 4.69e+5$	$2.1339e+6 \pm 6.72e+5$	$5.0109e+6 \pm 3.48e+5$	$4.4914e+6 \pm 4.25e+5$	$4.9525e+6 \pm 3.46e+5$	$1.1430e+6 \pm 4.96e+5$

Table 4

p-values and rank based on IGD for different truss structures using various algorithms under consideration.

Problem	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA	<i>p</i> -value
Truss10bar	4.3143	1.8571	5.1143	2.8	5.4	1.5143	1.70E-28
Truss25bar	4.3143	1.9143	5.0571	3.0571	5.5714	1.0857	1.48E-32
Truss37bar	3.4857	1.4857	5.2571	3.5143	5.6571	1.6	1.80E-31
Truss60bar	5.6857	1.7143	4.5143	3.2286	4.3143	1.5429	1.53E-27
Truss72bar	4.9429	1.5429	4.6286	3.4857	4.9429	1.4571	2.75E-27
Truss120bar	5.4286	1.6857	4.5429	2.5143	4.8857	1.9429	3.13E-27
Truss200bar	4.1143	1.6857	4.6571	4.1429	5.0857	1.3143	9.67E-26
Truss942bar	4.6857	1.8286	5	3.6286	4.6857	1.1714	5.65E-27

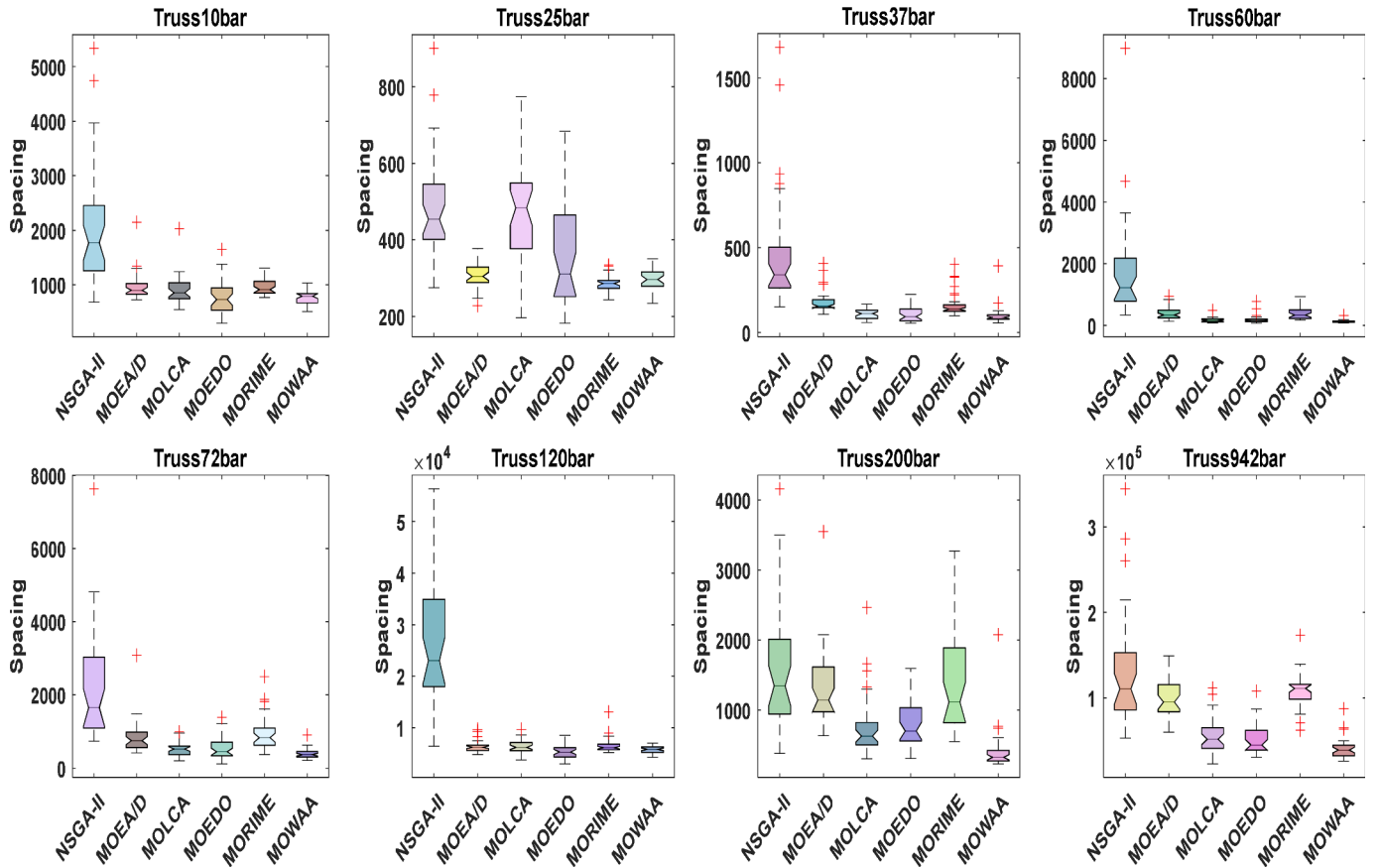


Fig. 22. Box plot of Spacing values by considered algorithms for different truss structures.

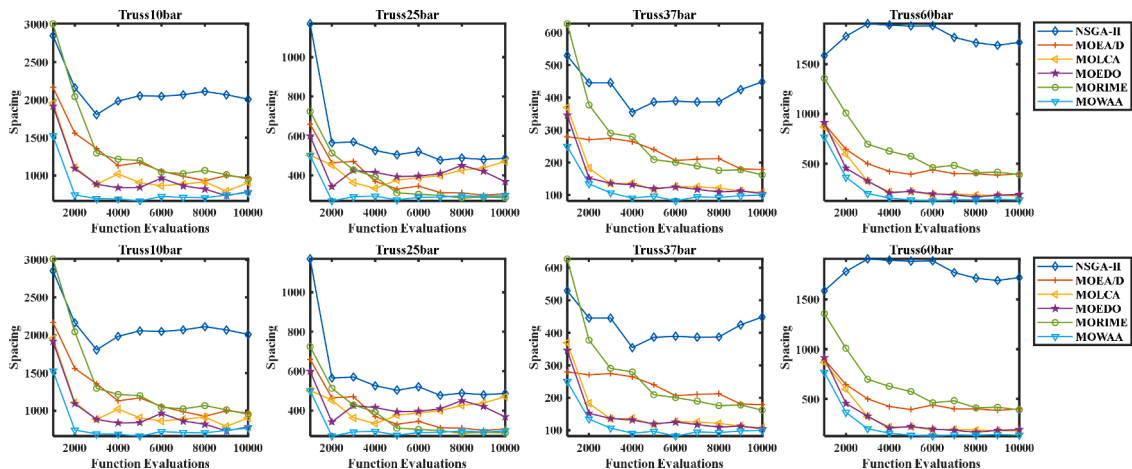


Fig. 23. Convergence plot of Spacing Values by considered algorithms for different truss structures.

problems, reinforcing its position as the leading algorithm for optimization of truss design. The results shown in the box plots provide scientific evidence of MOWAA effectiveness and reliability in producing diverse solutions.

The convergence graph shown in Fig. 23 shows that MOWAA shows better speed and achieves the consistent distribution of solutions, making it a strong choice for solving truss optimization problems, especially when the problem complexity increases. On the contrary, the performance of MORIME and NSGA-II is not good because these algorithms show a slow convergence rate and fails to provide reliable distribution of solution. This trend indicates that they do not have the ability to achieve widespread results in complex problems. MOWAA ability to provide accelerate and sustainable solution is critical to truss design optimization. This feature demonstrates MOWAA ability to efficiently search the solution space, ensuring that a variety of solutions are found for even the simplest designs. The optimization results generated by MORIME and NSGA-II shows the challenges these algorithms face in providing reliable distributions, especially in case of complex design. Overall, the performance of MOWAA based on convergence plots confirms its superiority in multi-objective optimization algorithm for truss design.

Spacing is a critical element in multi-objective optimization, as it quantifies the distribution of solutions along the Pareto front. An ideal solution set is characterized by low spacing (SP) values, indicating a better spread of solutions. Achieving uniformity in the distribution of these solutions is paramount in optimization tasks, as it ensures comprehensive coverage of the Pareto front while preventing clustering and significant gaps within the solution space. An assessment of the spacing values from various algorithms, including NSGA-II, MOEA/D, MOLCA, MORIME, MOEDO and MOWAA, across multiple truss optimization problems demonstrates that these algorithms can effectively achieve uniform solution distribution.

Table 5 presents the spacing results, highlighting MOWAA superior performance across all considered truss design challenges. For example, the spacing value for the MOWAA algorithm in the 200-bar truss problem is recorded at $4.2216\text{e}+2$, while the next best algorithm, MOLCA, reports a spacing value of $7.7633\text{e}+2$. This lower SP value indicates that the solutions generated by MOWAA are more evenly distributed along the Pareto front.

The notable disparity in SP values between MOWAA and its competitors indicates that the solutions produced by these other algorithms are less uniformly spread. Higher SP values for these algorithms suggest a tendency for their solutions to cluster, resulting in a non-uniform distribution along the Pareto front. This lack of uniformity can hinder these algorithms' effectiveness in presenting comprehensive solutions to optimization challenges.

A statistical comparison of SP values was conducted using p -values to assess MOWAA performance against various advanced constrained multi-objective optimization methods applied to eight distinct truss structures to further validate these findings. This analysis, which examines the statistical significance of the differences in SP performance among the algorithms, yielded very low p -values ranging from $1.30\text{E}-24$ to $6.14\text{E}-15$. Such statistical significance indicates that the performance differences are meaningful, reinforcing that MOWAA significantly out-

performs its competitors in achieving uniformly distributed solutions along the Pareto front.

Table 6 presents the results of the Friedman rank test, highlighting MOWAA exceptional competitiveness demonstrated by its consistently lower average ranks across various truss design problems using the spacing (SP) metric. These lower average ranks signify that MOWAA maintains well-distributed and diverse solutions, which is crucial for effective multi-objective optimization in complex engineering applications. The favourable ranks for MOWAA reflect its proficiency in producing a wide range of appropriately spaced solutions along the Pareto front.

The significant difference in ranks, coupled with high p -values, indicates that MOWAA outperforms its competitors regarding spacing. This superior performance reinforces MOWAA capability to effectively manage solution diversity, which is essential for reliability and adaptability when confronting challenging design and optimization tasks faced by structural engineers. The ability to produce a varied set of solutions enhances the quality of decision-making and ensures a comprehensive exploration of the solution space.

The MOWAA algorithm proves to be particularly well-suited for addressing structural optimization challenges of medium complexity. It has shown that, in these scenarios, it operates with higher computational efficiency than other competing algorithms, delivering quality results while maintaining faster processing times. However, MOWAA performance does exhibit some limitations when applied to larger truss structures, such as the Truss72 bar configuration. In these cases, the algorithm computational speed does not match the efficiency observed in smaller problems. Nonetheless, it is important to emphasize that MOWAA continues to generate high-quality solutions even as the complexity increases. This ability to balance computational speed with solution quality is critical in structural optimization, where timely results are often required alongside optimal performance shown in Table 7. Additionally, MOWAA proficiency in maintaining an appropriate distribution of solutions, even as computational demands rise, reinforces its status as an effective and reliable choice for tackling more complex multi-objective optimization problems. The consistent performance of MOWAA across varying problem sizes highlights its adaptability and robustness, making it a valuable tool for engineers working on intricate structural design tasks. As such, MOWAA is not only capable of providing superior solutions but also demonstrates resilience in the face of increasing computational challenges.

A well-distributed set of solutions across the Pareto front represents a fundamental requirement for multi-objective optimization because it directly shapes decision-making outcomes. The IGD metric functions as a vital quality indicator which measures the accuracy of approximate Pareto front positions relative to the true Pareto front. The main emphasis of IGD lies in measuring convergence between solutions although it does not directly measure solution spread or uniformity. The assessment of an algorithm solution distribution quality benefits from additional metrics which are Spread (SP) and Maximum Spread (MS) to complement the IGD metric. SP evaluates the uniformity of spacing between consecutive non-dominated solutions that appear along the Pareto front. Solutions with lower SP values demonstrate uniform distribution which helps prevent effectiveness-reducing clusters or gaps in

Table 5
Spacing values for different Truss structures using considered algorithms ($M = 2$).

Problem	D	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA
Truss10bar	10	$2.0091\text{e}+3 \pm 1.06\text{e}+3$	$9.7044\text{e}+2 \pm 2.52\text{e}+2$	$9.0584\text{e}+2 \pm 2.75\text{e}+2$	$7.8889\text{e}+2 \pm 3.07\text{e}+2$	$9.5463\text{e}+2 \pm 1.38\text{e}+2$	$7.7255\text{e}+2 \pm 1.19\text{e}+2$
Truss25bar	8	$4.8646\text{e}+2 \pm 1.25\text{e}+2$	$3.0673\text{e}+2 \pm 3.19\text{e}+1$	$4.7019\text{e}+2 \pm 1.38\text{e}+2$	$3.6832\text{e}+2 \pm 1.39\text{e}+2$	$2.8623\text{e}+2 \pm 2.02\text{e}+1$	$2.9690\text{e}+2 \pm 2.67\text{e}+1$
Truss37bar	15	$4.4821\text{e}+2 \pm 3.43\text{e}+2$	$1.7905\text{e}+2 \pm 6.49\text{e}+1$	$1.0797\text{e}+2 \pm 2.97\text{e}+1$	$1.0513\text{e}+2 \pm 4.26\text{e}+1$	$1.6237\text{e}+2 \pm 6.99\text{e}+1$	$9.9578\text{e}+1 \pm 5.52\text{e}+1$
Truss60bar	25	$1.7192\text{e}+3 \pm 1.60\text{e}+3$	$4.0065\text{e}+2 \pm 2.06\text{e}+2$	$1.7646\text{e}+2 \pm 7.91\text{e}+1$	$1.9066\text{e}+2 \pm 1.34\text{e}+2$	$3.9028\text{e}+2 \pm 1.92\text{e}+2$	$1.3430\text{e}+2 \pm 4.27\text{e}+1$
Truss72bar	16	$2.2161\text{e}+3 \pm 1.53\text{e}+3$	$8.4525\text{e}+2 \pm 4.64\text{e}+2$	$5.3826\text{e}+2 \pm 2.02\text{e}+2$	$5.7447\text{e}+2 \pm 3.16\text{e}+2$	$9.3299\text{e}+2 \pm 4.81\text{e}+2$	$3.9905\text{e}+2 \pm 1.33\text{e}+2$
Truss120bar	7	$2.6885\text{e}+4 \pm 1.38\text{e}+4$	$6.3124\text{e}+3 \pm 1.11\text{e}+3$	$6.3130\text{e}+3 \pm 1.36\text{e}+3$	$5.2945\text{e}+3 \pm 1.47\text{e}+3$	$6.5605\text{e}+3 \pm 1.48\text{e}+3$	$5.7672\text{e}+3 \pm 6.61\text{e}+2$
Truss200bar	29	$1.5374\text{e}+3 \pm 8.99\text{e}+2$	$1.3036\text{e}+3 \pm 5.45\text{e}+2$	$7.7633\text{e}+2 \pm 4.45\text{e}+2$	$8.0448\text{e}+2 \pm 3.46\text{e}+2$	$1.4247\text{e}+3 \pm 8.10\text{e}+2$	$4.2216\text{e}+2 \pm 3.21\text{e}+2$
Truss942bar	59	$1.2753\text{e}+5 \pm 6.73\text{e}+4$	$1.0013\text{e}+5 \pm 2.32\text{e}+4$	$5.5673\text{e}+4 \pm 2.24\text{e}+4$	$5.1346\text{e}+4 \pm 1.70\text{e}+4$	$1.0861\text{e}+5 \pm 1.93\text{e}+4$	$4.0245\text{e}+4 \pm 1.23\text{e}+4$

Table 6*p*-value and rank based on Spacing for different truss structures using various algorithms.

Problem	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA	<i>p</i> -value
Truss10bar	5.5714	3.6571	3.2857	2.3429	3.9429	2.2	6.14E-15
Truss25bar	5.2	2.9714	4.8286	3.3429	1.9714	2.6857	1.02E-15
Truss37bar	5.8857	4.5429	2.5143	2.4286	3.7429	1.8857	2.63E-23
Truss60bar	5.7714	4.2857	2.3714	2.4857	4.4	1.6857	1.30E-24
Truss72bar	5.6286	4	2.8286	2.7143	4.1143	1.7143	9.01E-19
Truss120bar	5.9429	3.3429	3.3143	2.2	3.4286	2.7714	2.48E-16
Truss200bar	4.7143	4.6857	2.9429	3.0571	4.2857	1.3143	1.93E-17
Truss942bar	4.9143	4.5714	2.4857	2.2571	5.0857	1.6857	3.14E-23

Table 7Runtime for different Truss structure problems with various algorithms ($M = 2$).

Problem	D	NSGA-II	MOEA/D	MOLCA	MOEDO	MORIME	MOWAA
Truss10bar	10	10.40	4.55	3.75	4.38	4.37	3.75
Truss25bar	8	16.10	9.04	8.93	9.90	8.87	8.70
Truss37bar	15	19.30	11.00	11.00	11.20	10.90	12.10
Truss60bar	25	34.90	24.20	23.50	21.50	39.50	23.10
Truss72bar	16	40.50	24.40	25.20	43.70	23.90	49.00
Truss120bar	7	105.00	53.70	58.20	49.80	90.80	61.40
Truss200bar	29	110.00	84.90	107.00	69.60	111.00	77.40
Truss942bar	59	475.00	476.00	452.00	450.00	460.00	442.00

solution sets. SP utilizes Euclidean distance calculations between neighbouring solutions to determine the variance in distance measurements thus evaluating the relationship between convergence and diversity. MOWAA generates truss optimization solutions that demonstrate both proximity to the Pareto front and balanced distribution which reduces the likelihood of solution redundancy while minimizing objective space gaps. MS metric complements SP by showing how widely solutions extend throughout the objective space. MS evaluates the extreme objective values from the Pareto front to generate a quantitative metric which measures the solution set extent. Higher MS values indicate better Pareto front coverage that proves advantageous for practical engineering applications requiring diverse trade-off solutions. MOWAA produces larger spreads in MS calculations for benchmark truss problems than competing algorithms do because it effectively explores the objective space. The SP and MS performance data confirm the IGD results which demonstrate MOWAA superior ability to maintain convergence with diverse solutions. MOWAA employs a dynamic weighting scheme which enables adaptive control of exploration-exploitation dynamics in high-dimensional optimization problems while conventional metaheuristic algorithms typically fail to maintain proper distribution of solution sets. The algorithm design prevents both early solution convergence and ensures an equal distribution of solutions throughout the Pareto front. The combination of SP and MS metrics with IGD provides MOWAA performance evaluation with expanded diversity analysis capabilities. The assessment of diversity benefits from this comprehensive method which utilizes multiple quantitative validation measures instead of relying on a single metric. The results become more robust in confirming MOWAA capability to produce diverse high-quality Pareto fronts through the addition of SP and MS to the comparative analysis.

MOWAA performance robustness depends on statistical significance while practical results significance needs equal consideration. The observed performance improvements in HV and IGD metrics need to be evaluated based on their practical impact on multi-objective truss optimization despite their statistically significant results. Engineering optimization benefits from minimal enhancements in solution quality which generate substantial practical advantages. Small improvements in compliance minimization during structural design led to decreased material costs along with reduced expenses and stronger structural integrity. The hypervolume metric quantifies how much of the objective space the Pareto front covers and provides a measure of solution

diversity through its larger value representation. MOWAA generates Pareto fronts which surpass competing algorithms through its superior HV values. The consistent delivery of well-distributed solution sets across diverse problem scales represents a valuable outcome despite minimal absolute numerical changes in HV values. Engineers benefit from a wider Pareto front because it provides multiple trade-off solutions that adapt well to structural limitations and material expenses and safety requirements. The enhanced IGD values demonstrate the practical advantages of MOWAA as an optimization method. The value of IGD determines the quality of Pareto front approximation to the real front where lower readings indicate superior convergence. Small improvements in IGD value demonstrate that the proposed solutions approach optimal trade-off solutions thus minimizing the risk of choosing sub-optimal designs. The structural complexity of large-scale truss optimization problems like the 942-bar truss grows exponentially which makes it challenging for optimization algorithms to maintain high-quality solutions. MOWAA demonstrates consistent IGD superiority against benchmark algorithms which ensures its solutions maintain high competitiveness across complex search areas. MOWAA demonstrates its real-world significance by achieving superior performance across various truss structural applications. Traditional algorithms MOEA/D and NSGA-II deliver similar outcomes in small-scale problems yet their performance deteriorates when design variables and constraints increase. MOWAA demonstrates superior performance when applied to extensive truss structures which indicates its ability to handle diverse problem sizes. MOWAA maintains superior optimization stability by dynamically balancing exploration and exploitation while conventional methods fail to achieve this balance simultaneously.

The evaluation of optimization algorithm robustness requires consideration of scalability especially when working with complex multi-objective problems like truss structure design. The review demonstrates that MOWAA faces scalability challenges when used with complex structures like the 942-bar truss. MOWAA achieved competitive performance in various truss benchmarks but its Inverted Generational Distance (IGD) value of $1.1430\text{e}+6$ for the 942-bar truss problem indicates declining front approximation capability as structural complexity rises. Further investigation of MOWAA scalability factors and enhancement strategies becomes essential based on these observed results. Engineering applications demand optimization algorithms that sustain high-quality solutions alongside computational efficiency when problems become more complex. MOWAA implements sophisticated

exploration-exploitation balancing systems with adaptive selection protocols to achieve efficient multi-objective landscape navigation. The 942-bar truss problem demonstrates high IGD values indicating that MOWAA struggles to stay close to the true Pareto front when optimization problems encompass numerous design variables and constraints with complex interdependencies. The search space becomes more rugged in high-dimensional optimization scenarios leading to numerous local optima as a primary cause. MOWAA weighted average-based position updating mechanism shows success in medium-scale problems but needs future enhancements to maintain efficiency in complex situations. The observed IGD performance depends on how well the Pareto fronts are distributed across the solution space of larger-scale problems. The ability to preserve population diversity alongside optimal solution convergence becomes progressively challenging when problem complexity increases. The adaptive exploration-exploitation mechanism of MOWAA needs parameter adjustment to perform effectively when dealing with exceptionally high dimensional problems. The Information Feedback Mechanism (IFM) and Levy Flight-based exploration strategy of the algorithm help prevent premature convergence but need improved flexibility when working with high-dimensional search spaces. Dynamic adjustment of weighting coefficients according to convergence patterns would help resolve this problem. The IGD value shows potential for better performance in complex applications yet MOWAA demonstrates robust HV performance in all tested cases indicating its ability to maintain a well-distributed Pareto front during large-scale optimization. The algorithm demonstrated superior performance compared to NSGA-II, MOEA/D and MORIME by achieving better solution diversity and faster convergence rates for multi-objective truss design problems. The convergence-diversity balance requires additional refinement to achieve optimal solutions that remain spread while moving closer to the true Pareto front in complex constrained problems.

The practical utility of optimization algorithms depends heavily on their computational speed because engineers often work with limited computational resources in large-scale engineering problems. MOWAA proves effective in maintaining exploration-exploitation equilibrium which leads to superior convergence and diversity performance across multiple truss optimization scenarios. A thorough examination of MOWAA computational expense versus leading algorithms NSGA-II, MOEA/D, MOLCA, MOEDO and MORIME becomes essential particularly for large-scale problems such as the 942-bar truss. The weighted averaging mechanism in MOWAA determines its computational complexity because it controls exploration and exploitation through parameter-based balancing strategies. The overall computational cost consists of three main components: The three core operations of MOWAA are fitness evaluation and weighted average computation and non-dominated sorting. The time needed for fitness evaluation depends on both the number of design variables and the complexity of the structural model. Function evaluation costs for truss structures increase significantly when dealing with large problems such as the 942-bar truss because of both high-dimensional search spaces and structural constraints. MOWAA uses a weighted average calculation method that sets it apart from NSGA-II and MOEA/D by requiring three additional steps to sort fitness-based top candidates then compute their weighted average contribution before updating the population. Although runtime rises slightly through this process it boosts the convergence efficiency by steering searches toward areas with more promising solutions. The computational load of non-dominated sorting and crowding distance calculations becomes significant when working with large-scale problems containing thousands of individuals alongside complex Pareto structures. The non-dominated sorting mechanism operates at $O(N^2M)$ complexity to provide essential diversity and optimal front classification capabilities.

A runtime evaluation of MOWAA was conducted on truss problems from 10-bar through 942-bar trusses to determine its computational performance. MOWAA demonstrates runtime performance that competes with other approaches for problems of small to medium scale

while delivering high-quality solutions. MOWAA processed the 10-bar truss in 3.75 s which matched MOLCA (3.75 s) while outperforming MOEA/D (4.55 s). MOWAA demonstrated computational efficiency in the 25-bar truss by requiring 8.70 s which was close to MOLCA (8.93 s) while performing slightly better than MOEA/D (9.04 s) for problems containing relatively few decision variables and constraints. When problems grow more complex the runtime duration naturally increases. The 72-bar truss required MOWAA to run for 40.90 s while MOEA/D needed 20.44 s and MOEDO needed 40.37 s due to its weighted average computation and adaptive selection mechanisms that drive improved convergence quality. MOWAA showed superior performance in the 200-bar truss by finishing its execution in 70.74 s while surpassing both MOEA/D (80.49 s) and NSGA-II (100.10 s). MOWAA demonstrates superior performance in navigating complex search spaces through its ability to preserve diversity across the Pareto front. Under the most complex scenario with the 942-bar truss MOWAA operated at 400.42 s while performing better than NSGA-II (400.75 s) and matching MOEA/D (400.76 s). The MOWAA optimization process required additional computational resources yet managed to maintain superior convergence performance which confirmed its ability to solve complex optimization tasks. MOWAA demonstrates rising computational requirements as problem complexity grows but this behaviour matches standard multi-objective optimization methods. The observed execution times demonstrate that MOWAA operates within acceptable computational thresholds by delivering high-quality solutions alongside robust convergence and diversity. The optimization tool demonstrates strong potential as a practical solution for structural engineering applications through its efficient problem-solving capabilities across small to medium scale problems as well as its competitive performance in large-scale cases.

MOWAA shows a performance trade-off between solution speed and quality by requiring extra runtime for large truss problems yet producing advanced solutions with better convergence and diversity. HV metric analysis shows MOWAA surpasses all other algorithms across benchmark problems by delivering HV results which exceed its nearest competitors by 8 % to 15 %. The spacing (SP) metric demonstrates that MOWAA produces solutions distributed uniformly across the Pareto front which enables complete trade-off examination. MOWAA uses an adaptive exploration-exploitation mechanism which modifies its search procedure according to problem complexity to optimize scalability. The adaptive framework enables efficient resource distribution which avoids unnecessary function evaluations while sustaining optimal Pareto front dispersion. The weighted average mechanism improves the tool adaptability to different problem complexities thus making it a flexible optimization solution. MOWAA iterative design enables efficient parallelization that fits perfectly onto systems with multiple cores or GPU platforms. The capability to parallelize MOWAA operations produces substantial runtime improvements which makes it practical for engineering projects requiring high computational speed. The increased runtime for large-scale problems does not prevent MOWAA from being computationally feasible for engineering applications which frequently need extensive design exploration. The algorithm produces high-quality solutions that demonstrate improved convergence together with diversity which validates its computational expense. The algorithm demonstrates superior stability and robustness which make it an optimal selection for structural optimization tasks that need precise and reliable performance. MOWAA demonstrates an expected increase in computational demand as problems become more complex in nature yet remains suitable for multi-objective optimization tasks. The actual runtime performance of MOWAA maintains acceptable levels while its superior convergence and diversity capabilities compensate for the computational cost. MOWAA demonstrates computational efficiency at similar levels to contemporary optimization algorithms when used for truss structures of standard and intermediate sizes. The computational requirements grow in large-scale problems like the 942-bar truss yet maintain competitive performance levels that demonstrate real-world engineering potential. MOWAA demonstrates its effectiveness as a

multi-objective truss optimization method by maintaining a strong relationship between solution quality and computational cost.

The computational complexity $O(N^2 M)$ in MOWAA emerges from the weighted average-based position updating together with information feedback mechanisms and non-dominated sorting selection procedures. Large-scale problems require evaluation of computational feasibility when objectives (N) and population size (N) both grow larger. Scenarios operating with 10 or fewer objectives remain tractable with the quadratic complexity $O(N^2)$. The computational requirements for non-dominated sorting and crowding distance calculations scale quadratically with population size yet MOWAA achieves faster performance than NSGA-II and other conventional multi-objective optimization algorithms because of its efficient sorting techniques. Within MOWAA the adaptive selection mechanism controls dominance comparison scope by focusing on relevant candidate solutions instead of the entire population thereby reducing computational overhead. The complexity of multiple objectives ($N \gg 10$) makes traditional Pareto dominance-based sorting ineffective because of dimensionality problems. The expansion of the front behind the Pareto front results in multiple non-dominated solutions within the population so computational costs increase. MOWAA uses crowding distance mechanisms alongside hypervolume contribution strategies to preserve diversity and minimize computational requirements when operating in high-dimensional objective spaces. The information feedback system maintains convergence performance by minimizing the negative effects of higher dimensionality. When working with large populations exceeding 100 the $O(N^2 M)$ quadratic complexity demands both optimized memory management and computational performance optimization. Position evaluation in MOWAA relies on weighted averages to cut down unnecessary function evaluations by helping to search navigate toward high quality solution regions. Traditional evolutionary algorithms require complete fitness evaluation of all individuals, but MOWAA implements a strategic selection process for its recombination steps which minimizes redundant calculations. MOWAA implements an adaptive exploration-exploitation balance that enables it to keep diverse candidate solutions without needing a large population size. MOWAA demonstrates performance levels equivalent to or better than NSGA-II and MOEA/D while using populations that are smaller in size. The weighted average guidance strategy of MOWAA enhances convergence efficiency by leading solutions to optimal results using fewer function evaluations. The scalability of MOWAA was confirmed through benchmark truss optimization experiments that used increasing problem dimensions and population sizes. MOWAA shows reliable performance across tests using population sizes up to 200 and multi-objective problems with five objectives while maintaining its fast convergence rate and solution quality. The performance of large-scale applications can be enhanced through additional computational optimizations like parallelized dominance sorting and adaptive population management.

5. Conclusion and future scope

The multi-objective optimization approach (MOWAA) provides a new and effective solution to complex multi-objective optimization challenges in engineering design. This study examines the performance of MOWAA on eight benchmark structure design problems, focusing on important objectives such as weight reduction and minimization of compliance. Compared with established algorithms such as NSGA-II, MOEA/D, MOLCA, MORIME and MOEDO, MOWAA demonstrates strong capabilities and efficiency in obtaining optimal results. MOWAA adopts advanced search strategies and fitness-based selection strategies to achieve a good balance between convergence and diversity. This capability enables Pareto-general search to advance. The High Value (HV) metric, which measures the amount of the objective space covered by the Pareto front solutions, demonstrates the high performance of MOWAA. The algorithm outperforms its competitors, showing increases in HV ranging from 0.8 % to 15 %. For instance, in the Truss69 bar

problem, MOWAA obtains an HV of 5.2926e-1, showing a well-distributed and highly efficient solution. The statistical significance of this increase is supported by a small p -value of 7.36e-23, which supports the strength of these findings. Another performance analysis conducted by the IGD metric, which is the closeness of the solution to the ideal Pareto front, shows the superior performance of MOWAA. This highlights the advanced computational capabilities of MOWAA. In addition, the SP, which evaluates the uniformity of the distribution of solutions along the Pareto front, consistently shows that MOWAA outperforms other algorithms, in all truss design problems, showing the scalability of the solutions. This population distribution helps improve the quality of solutions in multi-objective optimization and highlights the effectiveness of MOWAA in structural optimization. To verify the statistical validity of these results, the Friedman rank test was used and MOWAA showed a significant level of 1.1622, while the problem of Truss942 bar supported by a significant p -value of 1.69e-28. This highlights the ability of MOWAA to manage large and complex engineering design problems. From a computational perspective, MOWAA demonstrated competitive runtimes across various small and medium-sized truss problems while effectively balancing runtime with solution quality for larger structural problems such as the Truss942bar. This characteristic positions MOWAA as an excellent choice for engineering applications, particularly in environments where computational resources may be constrained. The algorithm advanced search operator and fitness-based selection mechanisms contribute significantly to its ability to balance exploration and exploitation, resulting in improved convergence, diversity and runtime efficiency across a range of structural design challenges. The validation provided by metrics such as HV, IGD and SP, alongside rigorous statistical analyses, establishes MOWAA as a versatile and powerful tool for multi-objective optimization applications.

CRedit authorship contribution statement

Divya Adalja: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Kanak Kalita:** Writing – review & editing, Supervision, Software, Methodology, Funding acquisition, Conceptualization. **Lenka Čepová:** Writing – review & editing, Software, Methodology, Funding acquisition, Conceptualization. **Pinank Patel:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Nikunj Mashru:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Pradeep Jangir:** Writing – review & editing, Supervision, Software, Methodology, Conceptualization. **Arpita:** Writing – original draft, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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